What a Robot Can Measure, It Can Undo: Separating Invertible Nuisances from Policy Failures at Deployment

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Abstract

A manipulation policy that works in the lab fails when the camera is moved, the arm starts somewhere new, or the light changes. Some of those failures are the policy's; others are only a change of viewpoint or pose that could be undone before the policy sees it. We ask which is which before any rollout and without the policy. The instrument is the robot's own body: encoders and a kinematic model say where its silhouette must appear, so the robot can measure the nuisance, estimate how much of the training view undoing it restores, and undo it for a frozen policy. On LIBERO-Plus with three policy families, a policy-free recoverability score, frozen before its test, predicts which failures the correction repairs on two held-out families (AUROC 0.81 and 0.68). Its value is inside a perturbation category, where the category label and any category-level router are at chance by construction: there it ranks repairability at 0.85 and 0.72 (pooled 0.74 [0.64, 0.87]); found post hoc, it replicates in a pre-registered test on unseen OpenVLA tasks (0.66, $p = 0.036$). A deployable variant that takes every camera input from the robot's own measurement and gates on its calibration residual does as well as the registered score (0.84 and 0.69); the residual alone flags every held-out calibration failure. With simulator masks and depth, a proprioceptive reset repairs 47 of 47 joint-offset failures on three untuned suites, and fitting the camera to the robot's silhouette repairs 61 of 100 camera failures (exact McNemar $p = 1.3 \times 10^{-6}$). An interventional ladder repairs 38 of the 39 camera failures that remain with the true pose (17) or the nominal observation (21), and an action-level tolerance of about 1 cm sets the target for real sensing: dense refinement on learned masks, with no depth sensor, brings the median calibration error from 13.2 cm to 1.65 cm, and a residual certificate tells the robot when its measurement cannot be trusted. All policy results are in simulation.

1 Introduction

Vision-language-action (VLA) policies are evaluated under one condition and deployed under another. LIBERO-Plus (Fei et al., 2025) made the gap concrete: moving the camera or the robot's initial pose takes policies that solve a task nine times in ten to below three in ten. The response has been to make the policy robust, or to wrap it: restore the corrupted stream (Yan & Shen, 2026; Guarnizo Orjuela et al., 2026; Hancock et al., 2025), re-render the view (Heo et al., 2026; Gu et al., 2026; Park & Kim, 2026), or predict the camera inside the policy (Li et al., 2026).

This paper asks a prior question. A deployment shift is either a nuisance, a change to how the scene is observed that leaves the task untouched and could in principle be inverted, or it changes the task. Benchmarks label perturbations by category, and wrappers route corrections by that label. The label does not tell you what is fixable: a camera that orbits 20 degrees is recoverable and one turned 35 degrees away is not, and both are "camera" (Fig. 1). What decides it is whether the robot can measure the change and whether undoing it restores what the policy was trained on. Both can be estimated before the policy acts.

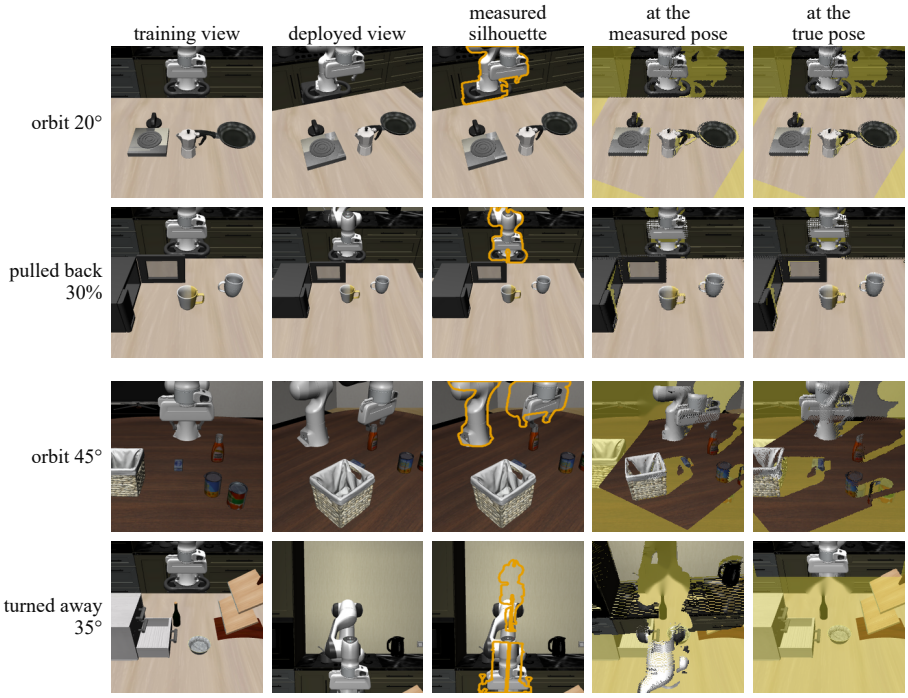


Figure 1: Four deployed cameras (OFT tasks; policy-free renders). Columns: training view, deployed view, the silhouette the robot fits its camera to from three probe poses (orange), the frame re-rendered into the training camera at the measured pose (yellow: training-view pixels the deployed camera cannot see, giving the coverage), and at the true pose. Rows 1-2 (recoverable; orbit 20 degrees, pulled back 30%): measured within 0.1 and 1.1 cm, coverage 0.77 and 0.98. Row 3 (not identified; orbit 45 degrees): the measurement fails (24.7 cm; the outline misses the arm), so the coverage it reports (0.53) is not the view; at the true pose coverage is 0.64 and the view returns. Row 4 (not invertible; turned away 35 degrees): the measurement also fails (98 cm), self-measured coverage reads 0.48, and even the true pose covers 0.23 of the training view.

The body as the instrument. A robot carries a calibrated object everywhere it goes: itself. Encoders give its pose; a kinematic model and a renderer give where its silhouette must appear for any camera pose. We use it to measure the nuisance, estimate how much of the training view undoing it restores, and undo it, with a frozen policy; a deployable variant of the score takes every camera input from the robot's own measurement (Sec. 4.1).

Contributions. (1) A policy-free recoverability score, frozen before its held-out tests, that transfers to two policy families it never saw and ranks repairability within a perturbation category, where labels and category routers cannot (0.74 pooled over the held-out families; Sec. 4.1). (2) Correction from self-measurement for a frozen policy: silhouette calibration, reprojection and stream restoration, with a proprioceptive reset that repairs every joint-offset failure on untuned suites and a motion-free silhouette check† that avoids measuring where nothing moved (Sec. 4.2, 4.3). (3) Where camera failures live: an interventional ladder in which the measured pose, the true pose or the nominal observation repairs 99 of 100 OpenVLA-OFT (OFT) camera failures, so few are the policy's, and an action-level calibration tolerance of about a centimetre† that real sensing must meet (Sec. 4.4, 4.5).

2 Related work

Wrappers around frozen VLAs. Restoring the Right Stream (Yan & Shen, 2026) is the closest work: a training-free per-stream restoration on the same benchmark, routed by stream identity. BYOVLA (Hancock et al., 2025) edits task-irrelevant image regions, CRT

(Guarnizo Orjuela et al., 2026) restores corrupted inputs, and ProbeAct (Zhang et al., 2026) repairs actions. We differ in predicting, before any rollout, which failures a correction can repair; routing by category and by the exact benchmark condition are our baselines (Sec. 4.1). Viewpoint. AnyCamVLA (Heo et al., 2026) re-renders the test view for a frozen VLA with a pose from fiducial markers; our reprojection is that method with a different pose source, and with the simulator's true pose in place of ours (Sec. 4.4) the same reprojection repairs 78 of 100 camera failures against our 61. VistaBot (Gu et al., 2026) and GS-VLA (Park & Kim, 2026) learn the synthesis; CamVLA (Li et al., 2026) regresses the camera inside a retrained policy; Li et al. (2025) trace viewpoint failures to spatial rather than physical modelling and adapt features. We do not claim reprojection. Robot as calibration target. Render-and-compare robot pose estimation (Lee et al., 2020; Labbé et al., 2021; Lu et al., 2023; Bilić et al., 2024) and mask-based hand-eye calibration (Chen et al., 2023; Hong et al., 2024) are established, DROID (Khazatsky et al., 2024) has been calibrated offline from robot masks (Xie et al., 2025; Irshad et al., 2025), and the conditioning of calibration Jacobians as an observability measure is classical (Borm & Menq, 1991; Nahvi & Hollerbach, 1996). Our silhouette calibration is in the EasyHeC and CtRNet family; the correction pipeline is therefore assembled from known parts, and what we add is the diagnosis it enables and the use of the observability measure at deployment, relative to the training view, as the first factor of a repairability score. Predicting failure. RoboART (Majumdar et al., 2025) predicts degradation under visual shift with the policy's own embedding; runtime monitors (Agia et al., 2024; Xu et al., 2025; Gu et al., 2025; Römer et al., 2025; Huang et al., 2026; Yang et al., 2026) read the actions a policy emits. Concurrently, VLA-Scope (Zhu et al., 2026) predicts failure under shift from the inputs, actions and execution history of a rollout, and Mahato & Ren (2026) probe OpenVLA's activations, noting that retrospective separability need not give quiet online warnings. Both need the policy and target failure, not whether a correction repairs it, so their numbers are not comparable with ours. Task-relevant shift (Farid et al., 2021), identifiability for test-time adaptation (Jhawar & Wang, 2026), selective adaptation (Jiang et al., 2026) and learned canonicalisation (Kaba et al., 2023) are conceptual ancestors; error amplification in closed-loop imitation (Block et al., 2024) and small-sample policy evaluation (Vincent et al., 2024) frame our statistics. We predict repairability, without the policy.

3 Measure, predict, undo, check

Let $o = g(s;\eta)$ be the observation of state s under nuisance parameters η (camera pose, initial joint configuration, illumination, sensor noise), η_0 the training setup, and π a frozen policy. A nuisance is invertible at η if a map T_η, computable from the robot's own measurements, gives $T_\eta(g(s;\eta)) \approx g(s;\eta_0)$ on the states the task visits. Fig. 1 draws the loop; every quantity below is defined by an explicit equation in Appendix B.1.

Measure. The robot records self-observations z that need no knowledge of the scene: joint readings, the silhouette of its arm at a few end-effector poses it visits deliberately, the colour statistics of its own links, and the noise level of its own pixels. The camera pose $\hat{\eta}_c$ minimises a silhouette chamfer plus a size term over the visited poses (coarse orbit grid, cross-entropy refinement, depth ICP) and is accepted only if it beats the training pose by a fixed factor (Eq. 13); light and noise are an affine colour fit and a noise floor on the robot's own pixels.

Predict. The score and its first factor are (Eq. 6)

$$\text{recover}(\eta) = \text{ident}(\eta) \times \text{invert}(\eta) \in [0,1], \qquad \text{ident}(\eta) = \min\Big(1, \frac{\sigma_{\min}(J_\phi(\eta))}{\sigma_{\min}(J_\phi(\eta_0))}\Big), \tag{1}$$

where J_ϕ is the finite-difference Jacobian of z with respect to the condition's own parameter block, columns in fixed steps (1 cm, 1° for the camera), and η_0 the same base task unperturbed: does the body still respond to the nuisance as it did in the training setup? In the registered score ident is computed from the simulator at the true nuisance (an oracle generative model of the self-observation: policy-free, not self-measured). The deployable score† evaluates the camera Jacobian at the self-measured pose $\hat{\eta}_c$ instead and multiplies by $\mathbb{1}[\rho \le \tau]$, where ρ is the calibration's own final silhouette residual: by Prop. 1 the two ident agree only where the calibration succeeded, and ρ says where it did. invert is per channel

and self-measured: for a camera, the fraction of the 256^2 training view that the reprojection from $\hat{\eta}_c$ fills on the first frame after the measurement (coverage c); for light, a split-half affine-invariant shading statistic of the robot's links; for noise, the share of the added noise level the denoiser removes on the robot's pixels; 1 for joint offsets; 0 for scene content, which has no inversion (App. B.1). Every constant was frozen and logged before the first episode of either held-out family.

Undo. Joint offset: servo to the training pose. Camera: reproject into the training camera with $\hat{\eta}_c$ and depth, z-buffered, holes filled from a stored training-view reference. Light: invert the affine fit unless the shading statistic says the change is not affine. Noise: denoise at the measured level. No weights change. The evaluated wrapper produces every headline number. Check (logged only). A self-check warps the robot's observed silhouette with the same pose and depth and compares it with where its kinematics place it in the training view (F1); the evaluated wrapper does not act on it.

3.1 What can be proved

Four facts carry the method (statements, conditions and proofs in Appendix B); for each we name the quantity it predicts and whether the prediction held. $z(q, \eta)$ is the self-observation at joint configuration q, J its Jacobian stacked over the chosen poses $q_1, \ldots, q_K$ in the d_η nuisance parameters ($d_\eta = 6$ for a camera), and $s^2_{\max}$ the largest eigenvalue of the sensing covariance Σ. (i) Measurement error (Props. 3, 8): in a local linear model with Gaussian sensing an efficient estimate has expected squared pose error between $\sigma_{\min}(\tilde{J})^{-2}$ and $d_\eta\, \sigma_{\min}(\tilde{J})^{-2}$, $\tilde{J} = \Sigma^{-1/2}J$, and adding a pose never lowers $\sigma_{\min}$; by a Markov bound (on the squared error), $\Pr(E_1) \geq 1 - d_\eta s^2_{\max}/(\text{ident}^2 \sigma_{\min}(J_\phi(\eta_0))^2 t^2)$ for the event E_1 that the pose error is at most t (Prop. 11). Predicted: a log-log slope of -1 of calibration error on $\sigma_{\min}$. Measured: -0.84 [-1.60, -0.08] at the probe reach the method uses; weaker at half reach, and the slope is sensitive to the spectrum's shape (Prop. 7). (ii) Coverage (Props. 4, 12): the corrected frame's error is exactly $(1-c)e_{\text{hole}} + c\, e_V$; the lower bound $(1-c)\bar{e}^\star$ is informative only without a stored fill (Fig. 5), and the coverage-repair relation (Fig. 2b) is empirical. (iii) Motion-free test (Props. 5, 18): at one rest pose, a silhouette residual above the largest one that bounded sensing and encoder error can produce at η_0 implies $\eta \neq \eta_0$; it separates every logged unit (Sec. 4.3). (iv) Self-measured identifiability (Prop. 1): ident at the measured pose differs from ident at the true one by at most $L_J\|\hat{\eta} - \eta\|/\sigma_{\min}(J(\eta_0))$, so it is trustworthy exactly where the calibration succeeded; the residual gate (Sec. 4.1) supplies that condition. Held: median difference 0.000 on the 40 units calibrated within 2 cm.

4 Experiments

Setup. LIBERO-Plus (Fei et al., 2025), four suites. Policies: OFT (Kim et al., 2025) with its per-suite checkpoints (development family), and two held-out families on which no constant was chosen: $\pi_{0.5}$ (Physical Intelligence et al., 2025) and OpenVLA (Kim et al., 2024) fine-tuned on LIBERO-10 (its released checkpoint and action statistics), a weaker policy (raw success under shift: 0.19 over 210 LIBERO-Plus tasks; OFT 0.76; $\pi_{0.5}$ 0.60 on 120 of them). A unit is one (condition, base task, seed), run raw and wrapped; it is repaired if raw fails and wrapped succeeds, broken if the reverse, so $\text{SR}_w - \text{SR}_{\text{raw}} = (R-B)/N$ over N units. For prediction a unit is (base task, condition), kept if the policy solves the base task unperturbed and its perturbed raw episode fails (App. B.1). The category label is whether a wrapper routing by LIBERO-Plus category would apply a correction (binary); the category rate is the repair rate of the unit's category fit on OFT. The frame is the observation handed to the policy. The graded sweep applies 24 conditions (nominal; 7 camera, 6 light, 4 noise, 3 joint, 3 background) to 10 base tasks. Set A is the first 30 LIBERO-10 camera tasks and set B the next 30. Simulator segmentation and depth stand in for an RGB-D sensor and a robot segmenter (Sec. 4.4). Statistics. AUROC is the Mann-Whitney statistic with ties counted one half, positive class "repaired"; intervals are percentile bootstraps over units (2,000 resamples, seed 0) and, in brackets, over base tasks. Paired comparisons use the exact two-sided McNemar test on discordant units (Eq. 3). The primary registered tests

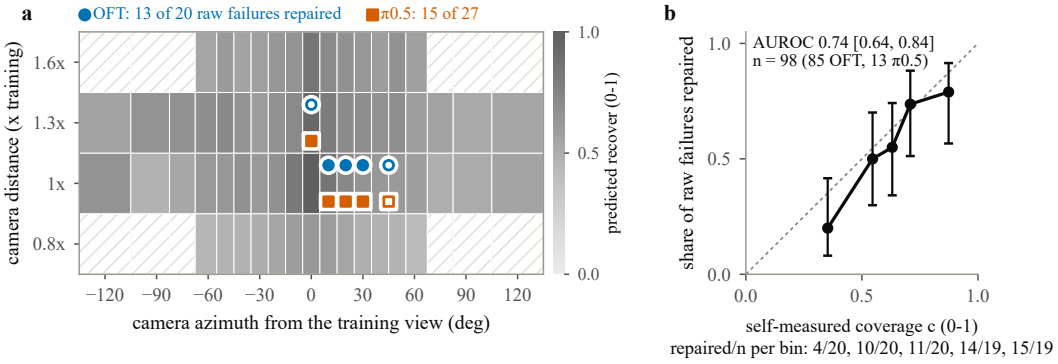


Figure 2: (a) The policy-free score recover over 88 camera placements × 3 base tasks (264 renders; mean over the three base tasks; grey ramp; hatched: not rendered) with the graded sweep's outcomes at its five standard camera placements (orbit 10, 20, 30, 45 degrees, pull-back 1.3; look-aways in Fig. 10), on base tasks each policy solves unperturbed (OFT circles in the upper third of a cell, $\pi_{0.5}$ squares in the lower third; filled: at least half of the raw failures at that placement repaired; OFT 13 of 20 raw failures repaired, $\pi_{0.5}$ 15 of 27). ident ≥ 0.9 on 199 of the 264 task-placements, so recover tracks coverage. (b) Every LIBERO-Plus camera task that logged coverage (four OFT suites and $\pi_{0.5}$; $n = 98$ raw failures: 85 OFT, 13 $\pi_{0.5}$; $\pi_{0.5}$ alone AUROC 0.63): observed repair against self-measured coverage in five equal-mass bins (Wilson 95%; dashed $y = x$); repair rises monotonically from 20% to 79%.

are the score AUROC on the two held-out families and the pooled OFT camera McNemar; the rest are secondary, with Holm-adjusted p in Table 6. Analyses designed after the main results are marked †; tests registered before their data existed are listed with their outcomes in Appendix A.

4.1 The recoverability score

Transfer. Frozen on OFT, the score predicts repair on both held-out families (Table 1), above the per-category rate fit on OFT on each (paired +0.11 [−0.03, 0.24] on $\pi_{0.5}$, +0.06 [−0.10, 0.22] on OpenVLA). Across categories most of its ranking comes from the self-measured coverage; the binary category label, nearly constant on these units, sits at 0.5 by construction. Repairability is largely a property of the nuisance, not the policy†: where OFT and $\pi_{0.5}$ both fail, their wrapped outcomes agree on 34 of 42 cells (Fisher $p = 1.3 \times 10^{-4}$), and there the score predicts $\pi_{0.5}$'s outcome better than OFT's own outcome does (0.94 against 0.81, +0.13 [0.05, 0.23]; OpenVLA +0.19 [0.12, 0.28]; cross_policy_transfer.py).

Within a category.† The question a router cannot answer is which failures of one kind are repairable: a camera orbit of 20 degrees is, a look-away of 35 is not, and both are "camera". Over pairs drawn from the same category any category-level predictor is exactly 0.5, and the score reaches 0.854 on $\pi_{0.5}$ and 0.720 on OpenVLA (pooled held-out 0.737 [0.64, 0.87], task-cluster bootstrap). Here the two factors combine: the score is above ident alone on every family (+0.15 pooled held-out, $p = 0.001$) and above invert alone on $\pi_{0.5}$ (+0.09, one-sided $p = 0.035$), and never below either (App. D). A finer lookup that knows the exact LIBERO-Plus condition, i.e. the true nuisance value the robot has to measure, reaches 0.644 [0.51, 0.81] (fit on OFT; score minus it +0.10, $p = 0.067$); within one condition it is not distinguishable from chance (0.54 [0.32, 0.74], 148 pairs), so what it ranks is mostly the nuisance's magnitude, measured rather than given. Pre-registered replication. With the deployable score (below) and four tests frozen before any row, 52 units on 3 unseen OpenVLA base tasks give within-category 0.663 ($p = 0.036$; pooled 0.655, $p = 0.007$; condition lookup 0.595), the residual flags all 10 calibration failures, and the deployable score reaches 0.667 against 0.615: all four pass (App. A). Look-away units, which the label calls recoverable, score 0.02 to 0.49, and 1 of 18 on $\pi_{0.5}$ is repaired.

Table 1: Predicting, without the policy, which raw failures the correction repairs (graded sweep). Units are (base task, condition) pairs, one rollout each in the sweep; positive = repaired (App. B.1). OFT 10 base tasks (9 solved unperturbed), $\pi_{0.5}$ 10 (9 solved), OpenVLA fine-tuned on LIBERO-10: 4 base tasks, 3 solved, 60 units (24 repaired); its task-cluster interval (App. D) rests on 3 clusters and is unreliable. All columns use the same units; the OFT category rate is fit in-sample. AUROC with 95% unit-bootstrap interval. Within category†: AUROC over (repaired, unrepaired) pairs from the same LIBERO-Plus category, where the label and the category rate are 0.5 by construction; there invert alone is 0.709, 0.761, 0.693 and ident alone 0.661, 0.656, 0.614.

	units (repaired)	score	ident only	invert only	category rate	category label	within† category
OFT (dev.)	65 (37)	0.823 [0.71, 0.92]	0.619	0.831	0.764	0.518	0.794
$\pi_{0.5}$ (held out)	72 (37)	0.809 [0.69, 0.91]	0.627	0.819	0.700	0.515	0.854
OpenVLA (held out)	60 (24)	0.675 [0.52, 0.82]	0.514	0.671	0.613	0.472	0.720

Table 2: Paired results: same tasks and seeds, raw and wrapped (n paired units; rec.: repaired of raw failures; broken: of raw successes). LIBERO-10 is the development suite, excluded from "47 of 47 on three untuned suites". The pooled OFT camera row is the primary registered test ($p = 1.3 \times 10^{-6}$); with the earlier v2 code on set A it is 58 repaired / 17 broken. All rows use simulator masks and depth except the last: learned masks and depth on set A (30 of the LIBERO-10 camera tasks), no better than raw.

policy	suite	joint offset				camera			
		n	raw → wrapped	rec.	broken	n	raw → wrapped	rec.	broken
OFT	Spatial	20	0.25 → 0.95	15/15	1/5	60	0.90 → 0.88	3/6	4/54
OFT	Object	20	0.20 → 1.00	16/16	0/4	60	0.42 → 0.75	26/35	6/25
OFT	Goal	20	0.20 → 1.00	16/16	0/4	60	0.67 → 0.78	12/20	5/40
OFT	LIBERO-10	30	0.33 → 0.90	18/20	1/10	60	0.35 → 0.63	20/39	3/21
OFT	pooled	90	0.26 → 0.96	65/67	2/23	240	0.58 → 0.76	61/100	18/140
$\pi_{0.5}$	LIBERO-10	30	0.37 → 0.90	16/19	0/11	30	0.20 → 0.37	7/24	2/6
OFT, learned	LIBERO-10					30	0.43 → 0.40	2/17	3/13

Coverage is the physical quantity. On every LIBERO-Plus camera task that logged it (98 raw failures), repair rises monotonically from 20% to 79% with self-measured coverage (AUROC 0.74 [0.64, 0.84]), above the shift magnitude that caused it (Fig. 2b; App. D). Coverage at a failed pose can still read high, so coverage gates inversion, not measurement.

Every camera input self-measured.† Rescoring all 70 camera units with the Jacobian at the self-calibrated pose, the two ident agree where the calibration is within 2 cm (median difference 0.000, 40 units) and the self-measured one is too optimistic where it fails (0.91 against 0.70, 30 units), as Prop. 1 allows. The calibration's own silhouette residual separates its successes ($\rho \leq 0.58$) from its failures ($\rho \geq 0.65$) on all 119 logged camera units; gated at τ fit on OFT alone, it flags all 44 held-out failures (one false alarm in 54), and the deployable score reaches 0.837 on $\pi_{0.5}$ and 0.691 on OpenVLA, at or above the registered oracle score (App. D). The robot knows both whether a shift is invertible and whether its measurement of it can be trusted. Pair decompositions, the registered coverage-gate rule and OpenVLA units repaired with no channel applied are in App. D.

4.2 Undoing what was measured

Joint offsets (Table 2). The servo returns the arm to the nominal start from proprioception and repairs 47 of 47 failures on the three untuned suites with 1 of 13 successes broken ($p = 3.5 \times 10^{-13}$), and 16 of 19 with none broken on $\pi_{0.5}$: the robot-initial-state category of LIBERO-Plus is recoverable by a proprioceptive reset, a benchmark finding rather than a correction algorithm. On the same tasks and seeds†, the reset solves more than placing the arm at the training start (30/30 against 22/30 on LIBERO-10, 59/60 against 32/60 on the others; $p \leq 0.04$ on each suite), so the reset is not limited by that base rate. Camera. Over

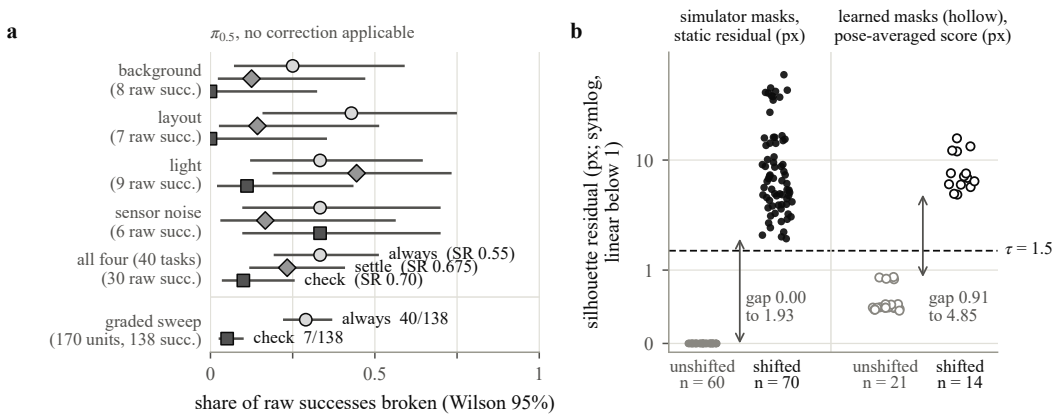


Figure 3: (a) $\pi_{0.5}$, 40 tasks in four non-camera categories with no correction applied (30 raw successes): share of raw successes broken (Wilson 95%) when the wrapper measures always (circle), measures then settles (diamond), or runs the motion-free check first (square); success rates 0.55, 0.675 and 0.70 (decomposed in App. D)†. Bottom: the 170-unit graded sweep (138 raw successes), where the check never fires and so equals never measuring (task-level $p = 0.031$ vs. measuring always, 0.047 vs. raw)†. (b) Filled: one-pose silhouette residual (Eq. 12), simulator masks, policy-free: 60 unshifted units (10 nominal, 50 lighting; largest 0.00; 0.020 over the 170 $\pi_{0.5}$ units) against 70 camera shifts (smallest 1.93). Hollow: learned masks, the pose-averaged score S (Eq. 14), a different statistic (21 unshifted OFT rows, largest 0.906; 14 shifted, smallest 4.85). $\tau = 1.5$ chosen post hoc.

the four suites the wrapper repairs 61 of 100 camera failures and breaks 18 of 140 successes ($p = 1.3\times10^{-6}$; excluding the development suite, 41 repaired / 15 broken, $p = 6.9\times10^{-4}$); it is significant alone on Object ($p = 5\times10^{-4}$) and LIBERO-10, and the gain follows need: on Spatial, whose checkpoint already survives 90% of shifts, it is neutral. On $\pi_{0.5}$ the registered single rollout does not resolve the camera channel (7 repaired, 2 broken, $p = 0.18$); over two further replicates† the trend is positive (21 and 8 over 90 episodes; exact task-level sign-flip $p = 0.12$). Set B baselines are in Table 5 (App. D): depth-free homography, generic fixes and test-time augmentation all lose to the wrapper.

4.3 The cost of measuring, and a test that removes it

Self-measurement moves the arm, and the held-out $\pi_{0.5}$ is sensitive to where its episode starts. Over the whole graded sweep (240 units) the measuring wrapper repairs 44 of 82 $\pi_{0.5}$ failures and breaks 45 of 158 successes, net zero: the probe motion hands the arm back 10 to 25 mm off and still moving, which changes 15 of 48 outcomes where nothing else was applied (0 of 20 on a same-seed rerun; App. D). Prop. 5 removes the motion†: the robot looks once at its own silhouette without moving and measures only if the residual exceeds $\tau = 1.5$ px; otherwise it settles (returns the arm to the episode's start and waits until it is still).

The residual separates the cases on every logged unit (Fig. 3b; App. B.4); with no error observed on 10 base-task clusters, the one-sided 95% upper limit on either error rate is 0.259 (Prop. 19). Between the tolerance (under 1° of orbit) and about 8° of orbit the check is not shown to fire (Lemma 2). On the 170 sweep units without a camera shift (138 raw successes, 32 raw failures), where the test must not fire and did not, the measuring wrapper breaks 40 and repairs 25; the motion-free variant breaks 7 and repairs 21, keeping 18 of the 25 repairs and adding 3 (Fig. 3a; vs. measuring always 41 wins / 12 losses, $p = 8.2\times10^{-5}$; vs. raw 21 / 7, $p = 0.013$; task-level† $p = 0.031$ and 0.047). Where it must decide† ($\pi_{0.5}$, 60 rollouts), it fires on all 30 camera tasks (smallest residual 4.04) and on none of 30 joint-offset tasks (largest 0.01), matching always-measuring on camera (7/24 repaired, 2/6 broken for both) and within noise on joint offsets (14/19 and 1/11 against 16/19 and 0/11; $p = 0.38$).

Table 3: Attribution ladder, OFT, set A (13 raw successes, 17 raw failures; the two failure rows differ by the non-monotone units, Prop. 20). On set A, no failure survives a perfect render (pooled over the five OFT sources, 1 of 100 does; Fig. 8); the 8 render-only units have the lowest coverage. Last column: learned masks and depth, not nested. *At the failed measured calibration ($n = 2$).

diagnoses	raw	measured pose –	true pose not identified	perfect render not invertible	learned sensing
success (of 30)	13	19	21	27	12
raw successes broken (of 13)		1	0	0	3
raw failures that succeed (of 17)		7	8	14	2
first variant that repairs (of 17)		7	2	8	
self-measured coverage, median		0.58	0.96*	0.44	

4.4 Where the deficit lives

When a corrected policy still fails, we find which part failed by intervention†: on the same task and seed, four nested variants each remove one more cause (raw; measured pose; the simulator's true pose with the same reprojection and training-view fill; a perfect render from the training camera), holding the measurement motion and fill fixed, and a raw failure is attributed to the first variant that repairs it. The rule is well posed for any outcomes, and a unit is misattributed with probability at most $\min(\beta^*, 3)f$, with β^* its noise-free bucket (the first rung that would repair it without rollout noise, 4 if none) and f a per-unit flip bound (Prop. 20); with observed buckets and $\hat{f}$, the outcome-flip rate between raw and wrapped rollouts on the OFT sweep units where the wrapper applied nothing (7 of 53; App. B.1), this is a plug-in estimate.

On set A (Table 3), OFT's raw camera failures are a failed measurement or a lost view. The perfect-render rung is the nominal observation, and mostly a base rate: on the same tasks and seeds the unperturbed view† solves 15 of the 21 OFT render-only units (set A 8 of 8, pairing across harnesses; the other suites 7 of 13, Goal 1 of 6), and we count the 6 it does not solve as single-rollout noise inside the misattribution budget of Prop. 20. Extending the upper rungs to every raw camera failure on all four suites, 100 OFT failures split into 61 first repaired at the measured pose, 17 only by the true pose, 21 only by a perfect render, and 1 by nothing; on the held-out $\pi_{0.5}$ 5 of 24 survive every rung (7, 3, 9 and 5; Fig. 8; both upper rungs pictured in Fig. 1). Under independent flips at the upper limit of the flip rate, at most 8 of the 100 belong to the policy (a plug-in estimate; bucket uncertainty in App. D). Only-true-pose units are gross calibration failures (OFT median 39.4 cm); only-render units have the lowest self-measured coverage. Without rollouts†, the dose at which the median opening action changes by 10% is about 1 to 2 cm and 0.5 to 1 degree in both families (Fig. 4; frames in Fig. 7); the knee moves to 0.5 and 4 cm at thresholds 0.05 and 0.20, and the sublinear response bounds it by the smallest crossing dose, at most 0.96 cm for OFT and 1.39 cm for $\pi_{0.5}$ (Prop. 17). In-rollout repair rates by calibration error show no knee at 1 cm (App. D). On 8 policy-free tasks self-calibration reaches 0.13 to 0.54 cm; during the evaluated OFT camera rollouts its median error is 1.6 cm (54% within 2 cm, $n = 219$ of 240 evaluated units; 21 set-B units did not log the error), and the measurement failures are the tail.

Sensing is the requirement. Learned masks and depth reach 12.6 cm on set A and do not improve on the raw policy (12 of 30 against 13; 19 with simulator sensing), and a monocular metric-depth network† does not help (11.4 cm, 8 of 30); the residual gate still flags all 29 of those fits beyond 2 cm. The fix is the pose, not depth†: on the learned masks of the same 30 sweeps, refining the pose from two starts and keeping the lower dense loss takes the median error from 13.2 to 1.65 cm (16 of 30 within 2 cm, from 1; policy-free, no depth sensor); 12 of the 14 beyond 2 cm are shifts of 44 cm or more, 6 of the 14 end within 3 cm, and the final loss flags them at AUROC 0.82. Pose quality is the lever†: on set B, refining the same silhouette pose EasyHeC-style against dense masks reaches a median 0.29 cm and solves 23

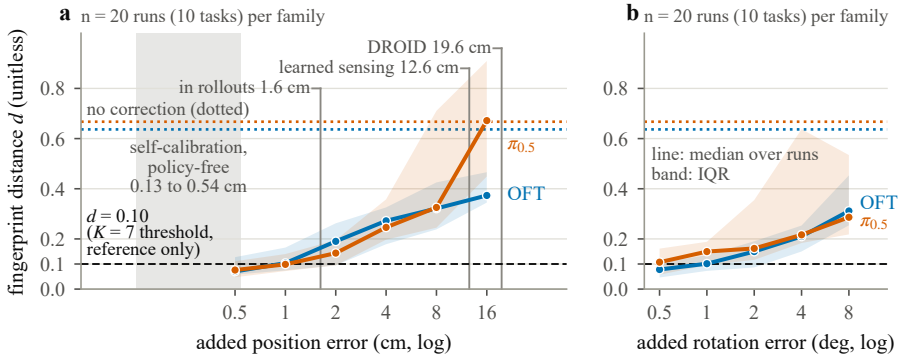


Figure 4: How exact the calibration must be, with no rollouts†. Fingerprint distance d (Eq. 15, $K = 1$: the first action chunk at the episode's untouched start state) between the opening actions at the true camera pose plus a known (a) position or (b) rotation error and at the true pose (reference: the true-pose corrected frame); median and IQR over $n = 20$ runs per family (orbit 10 and 20 degrees, coverage ≥ 0.68, 10 tasks each). Dashed: $d = 0.10$, the $K = 7$ threshold, drawn for reference only (knees in Sec. 4.4). Dotted: no correction. Grey band: policy-free self-calibration error on 8 tasks (0.13 cm with 7 probe poses to 0.54 cm with 1). Leaders: median calibration error in the evaluated OFT camera rollouts (1.6 cm, $n = 219$), with learned sensing (12.6 cm, $n = 30$), on real DROID images (19.6 cm, $n = 53$).

of 30 tasks against the search's 19 (4 wins, 0 losses, $p = 0.125$), and a fiducial pose with oracle corners solves 25 ($p = 0.031$).

4.5 What the policy's own actions say

Queried at seven end-effector positions the robot knows from its encoders, a policy's commanded motions should meet at a sub-goal if it understands the frame. They do not: the triangulation residual does not predict wrapped success (AUROC 0.32 to 0.56), and the fitted local field expands volume on 96-97% of units (App. D, Prop. 15). What the probes do carry is a fingerprint, the distance d (Eq. 15) between the seven probe actions under a frame and under the same task in its training setup: it predicts wrapped failure at 0.93 (OFT), 0.787 ($\pi_{0.5}$) and 0.643 (OpenVLA), against a registered bar of 0.80 (Table 7).

5 Limitations

Sensing. The main results use simulator segmentation and depth; with learned masks, refinement reaches a median 1.65 cm without rollouts, and its closed-loop test is pre-registered (App. A); the true pose repairs 78 of 100 camera failures against the measured pose's 61. Scope. All policy evaluation is in simulation on one benchmark with three policies, with no real-robot rollout; DROID tests only self-measurement on real images, where the residual gate is weaker (App. C, D). The held-out camera test on $\pi_{0.5}$ ($n = 30$ tasks) is not significant, the light channel on $\pi_{0.5}$ trends harmful (App. D), and the within-category and deployable-score results were found post hoc; their pre-registered replication rests on 52 units from 3 unseen OpenVLA base tasks (App. A). Design. The invert terms are per channel, and a calibration takes about a minute (pause-and-measure deployment).

6 Conclusion

A robot's body tells it, before any rollout and without the policy, which deployment failures are nuisances it can undo and whether its own measurement can be trusted; almost no remaining camera failure is the policy's. Dense refinement brings learned sensing to about the required centimetre, and a residual certificate says when it has not.

AI use statement

We used generative AI tools (large language model coding and writing assistants) to write and debug experiment code, to recompute and cross-check the reported statistics from the logged result rows, to search for related work, and to draft and edit text. A vision-language model is also part of one experiment, where it chooses robot mask regions on DROID images (Appendix C); there it is a method component. Every reported number was recomputed from the released result rows with the analysis scripts, and every reference was checked against its arXiv, proceedings or publisher record. We have reviewed all AI-assisted work and take responsibility for the final content of this paper, including text, claims and artifacts produced with the aid of generative AI.

Reproducibility statement

Every constant of the score, the fingerprint and the registered decision rules was written to a dated log before the data that tests it existed, and each registered test is reported against that entry, whatever its outcome (Appendix A). Definitions and proofs are in Appendix B. Code, the run notebooks and all per-episode results will be released.

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A Registered tests and their outcomes

Each test below was written to a dated log, with its constants and pass criteria, before any data that could test it existed. Development-family (OFT) outcomes are descriptive; held-out outcomes are the test.

The H-F2 values are the registered raw-frame statistic; they are not the "opening action only" row of Table 7, which is a post hoc ablation of the corrected-frame statistic.

test (registered)	rule	outcome
Recoverability score as a predictor (frozen 2026-09-17)	rank raw failures by recover; no numeric bar registered	descriptive; no pass criterion was registered: 0.809 [0.69, 0.91] ($\pi_{0.5}$), 0.675 [0.52, 0.82] (OpenVLA). Post hoc note†: both above the category label (0.515, 0.472)
Do-no-harm gate (2026-09-18)	veto a correction if the action changes more under benign jitters of the corrected frame than of the raw one	fails: over 386 paired gated / ungated units on the two held-out families it vetoed 49, saved 1 unit and broke 16 ($p = 3 \times 10^{-4}$); where it does not veto it reproduces the ungated variant exactly
Score-gated correction (2026-09-19)	correct the camera only if coverage ≥ 0.5	killed by its own criterion on rows unseen when frozen: prevented 2 breaks, gave up 3 repairs
Ray-concurrency probe (2026-09-19)	triangulate a sub-goal from 7 probe actions	fails on both families (AUROC 0.32 to 0.56); explained by Prop. 6
Action-field fingerprint, H-F1 to H-F5 (amendment, 2026-09-19, before any held-out probe)	H-F1 corrected AUROC ≥ 0.80; H-F2 raw AUROC ≥ 0.70; H-F3 keep raw on $\geq$ half the units a correction breaks; H-F4 keep the correction on $\geq 80\%$ of units it repairs; H-F5 selected success $\geq$ best fixed variant minus one unit	$\pi_{0.5}$: H-F3 and H-F5 pass (5 of 5; 0.86 against 0.81), H-F1 0.787, H-F2 0.684, H-F4 7 of 9 narrowly fail. OpenVLA: H-F1 0.643, H-F2 0.659, H-F4 3 of 13, H-F5 0.32 against 0.46 fail; H-F3 undetermined (one unit)
DROID self-calibration stop rule (written before the final calibration run)	stop if the median error on reliable references exceeds 30 cm	fired (32.1 cm); the later mask change reaching 19.6 cm is post hoc (Appendix C)
Replication on 6 unseen OpenVLA base tasks (frozen 2026-09-25, before any row; wrapper/prereg/)	deployable score (self-measured ident, residual gate $\tau = 0.547$): H1 within-category AUROC > 0.5 (within-category permutation, one-sided 0.05); H2 same, pooled with the 60 registered units; H3 residual flags calibration failure, AUROC ≥ 0.90; H4 AUROC within 0.05 of the registered score	all pass (3 of 6 tasks solved, 52 units, 22 repaired): H1 0.663, $p = 0.036$; H2 0.655, $p = 0.007$; H3 1.000 (21 units, 10 failures); H4 0.667 vs 0.615

B Definitions, propositions and proofs

B.1 Definitions of the measured quantities

Every quantity reported in the paper is one of the following, computed exactly as written; the constants are the frozen values of Appendix A.

Units and outcomes. An episode is a (policy, task, condition, variant, seed) tuple with binary outcome $Y \in \{0, 1\}$. Variants are raw (the frozen policy on the perturbed task) and wrapped (the same task and seed with the wrapper). When a unit is replicated R times its outcome is $\bar{Y} = R^{-1} \sum_{\ell} Y_{\ell}$ and it counts as a success iff $\bar{Y} \geq 1/2$. In the graded sweep a prediction unit is a (policy, base task, condition) with raw success on the unperturbed base task and raw failure on the perturbed one; its label is $y = \mathbf{1}[\text{wrapped succeeds}]$. The sweep has a single rollout per episode ($R = 1$; no (task, condition, variant) key repeats in its rows), so the replicate rule does not act there. The sweep uses the 24 standard conditions only; the boundary conditions are a separate analysis and are never pooled with them.

Repaired, broken, and the paired test. On a set of paired tasks with raw outcomes Y_i^{r} and wrapped outcomes Y_i^{w},

$$b = \#\{i : Y_i^{\mathrm{r}} = 0,\ Y_i^{\mathrm{w}} = 1\}, \qquad c = \#\{i : Y_i^{\mathrm{r}} = 1,\ Y_i^{\mathrm{w}} = 0\}, \tag{2}$$

"repaired" is reported as $b/\#\{i : Y_i^{\mathrm{r}} = 0\}$ and "broken" as $c/\#\{i : Y_i^{\mathrm{r}} = 1\}$. The exact two-sided McNemar test conditions on the $n = b + c$ discordant pairs, which under the null of no effect are Binomial$(n, 1/2)$:

$$p = \min\Big(1,\ 2 \sum_{k=0}^{\min(b,c)} \binom{n}{k} 2^{-n}\Big), \qquad p = 1 \text{ if } n = 0. \tag{3}$$

Concordant pairs do not enter. Example: joint offsets on the three untuned suites have $b = 47$, $c = 1$, so $p = 2(1 + 48)\,2^{-48} = 3.5 \times 10^{-13}$. Every "wins / losses" comparison between two variants is (2) with the two variants in place of raw and wrapped.

AUROC. For scores s_i on P positive and N negative units (units with an undefined score are dropped),

$$\widehat{A} = \frac{1}{PN} \sum_{i:y_i=1} \sum_{j:y_j=0} \Big(\mathbf{1}[s_i > s_j] + \tfrac{1}{2}\,\mathbf{1}[s_i = s_j] \Big), \tag{4}$$

the Mann-Whitney statistic with ties counted one half (equivalently, mid-ranks). Intervals are percentile bootstrap intervals, 2,000 resamples of units with replacement, seed 0, resamples with a single class discarded (none was discarded for Table 1, unit or task-cluster); task-cluster intervals resample base tasks with all their units. The category label is scored by (4) with $s \in \{0, 1\}$ (camera, light, noise, joint = 1; background = 0), so its AUROC is $(1 + \mathrm{TPR} - \mathrm{FPR})/2$ and a constant label within the units scores exactly 0.5. The category rate replaces s by the repair rate of the unit's category (or condition) on OFT.

Self-observations and ident. At the swept joint configurations $q_1, \ldots, q_K$ the robot records

$$z(\eta) = \Big[G_2 * M_k \ (k \le K),\ \ \tfrac{10}{255}\,\bar{c}_k \ (k \le K),\ \ 10\,q_k \ (k \le K),\ \ \tfrac{10}{255}\,\nu(o) \Big], \tag{5}$$

with M_k the robot mask at 96^2 pixels blurred by a Gaussian of standard deviation 2 pixels, $\bar{c}_k \in [0, 255]^3$ the mean colour of the robot's pixels in a 96^2 render, q_k in radians and ν the noise level defined below, evaluated on a 128^2 render of the current state with the condition's noise added (generator seed 0). For a condition of family ϕ with parameter block θ_ϕ (camera: 3 translations of 1 cm and 3 rotations of 1°; directional light: a 5° azimuth rotation and an elevation tilt that replaces the direction's vertical component d_z by $d_z \cos 5° - 0.087$ and renormalises, about 5°; other light: a gain of 1.1 on each diffuse channel; noise: +2 grey levels of noise standard deviation; joints: 0.02 rad per arm joint; table colour: +0.1 per RGB channel, clipped at 1; coloured room: one step of the single scalar that sets table colour and light tint, applied on top of the current state at 0.1 of the condition's strength), $J_\phi(\eta)$ has one column per block parameter, $J_\phi(\eta)_{:,i} = z(\eta + h_i e_i) - z(\eta)$ with the step h_i above (to first order $J_\phi = (\partial z/\partial \theta_\phi)\,W$ with $W = \mathrm{diag}(h)$, so $\sigma_{\min}$ is in these units), and

$$\mathrm{ident}(\eta) = \min\Big(1,\ \frac{\sigma_{\min}(J_\phi(\eta))}{\sigma_{\min}(J_\phi(\eta_0))}\Big), \qquad \mathrm{ident} = 0 \text{ if } \sigma_{\min}(J_\phi(\eta_0)) = 0. \tag{6}$$

The steps are the same at η and η_0, and the reference η_0 is the same block of the same base task in the training setup.

What ident is computed from†. The finite-difference columns of (6) perturb the simulator at the true nuisance: for a camera condition the six columns are z rendered at $(C + he_i, R)$ and $(C, \exp(h[e_i]_\times)R)$ with (C, R) the simulator's true deployed pose (block_deltas, which reads true_camera()); for light, table colour and joints the true model arrays are stepped; and the calibration $\hat{\eta}_c$ is computed only afterwards and enters only the coverage $\hat{c}$. So ident is policy-free but not self-measured: it uses an oracle generative model of the self-observation at the true nuisance. Of the two factors, only invert (coverage for a camera, and the shading and noise statistics, which are read from the deployed frame) is self-measured. A deployable ident evaluates the Jacobian at the self-measured nuisance, $\mathrm{ident}(\hat{\eta}) = \min(1, \sigma_{\min}(J(\hat{\eta}))/\sigma_{\min}(J(\eta_0)))$.

Proposition 1 (oracle and self-measured ident). Let $J(\cdot)$ be the step-scaled Jacobian (or its finite-difference secant) of the self-observation in the block's parameters, with $\eta \mapsto J(\eta)$ Lipschitz in operator norm on the segment $[\eta, \hat{\eta}]$, $\|J(\eta') - J(\eta'')\|_2 \le L_J \|\eta' - \eta''\|$, with η in step units (1 cm, 1°). Then

$$\big|\sigma_{\min}(J(\hat{\eta})) - \sigma_{\min}(J(\eta))\big| \le L_J\,\|\hat{\eta} - \eta\|, \qquad \big|\mathrm{ident}(\hat{\eta}) - \mathrm{ident}(\eta)\big| \le \frac{L_J\,\|\hat{\eta} - \eta\|}{\sigma_{\min}(J(\eta_0))}. \tag{7}$$

Proof. Weyl's inequality for singular values, $|\sigma_i(X) - \sigma_i(Y)| \le \|X - Y\|_2$ for every i, applied with $i = d_\eta$ to $X = J(\hat{\eta})$, $Y = J(\eta)$, and the Lipschitz condition give the first bound. The

map $x \mapsto \min(1, x/\sigma_{\min}(J(\eta_0)))$ is $1/\sigma_{\min}(J(\eta_0))$-Lipschitz (a composition of a linear map with the 1-Lipschitz $\min(1,\cdot)$), which gives the second. □

The bound is linear in the calibration error, so it is informative only where the calibration succeeds; at a gross failure (the 108.9 cm unit of Fig. 5) the two scores need not be related. Check. L_J is not logged; the sweep logs $\sigma_{\min}$ at the unperturbed camera and at orbits of 10, 20, 30 and 45° (and 15, 25, 38° on three tasks), whose logged true shifts give the chord in step units, $\|\Delta\eta\| = (\Delta_{\text{cm}}^2 + \Delta_{\text{deg}}^2)^{1/2}$. Over the 49 consecutive pairs on 10 tasks, $|\Delta\sigma_{\min}|/\|\Delta\eta\|$ has median 0.0071 and maximum 0.0237 per step unit; this is a secant of $\sigma_{\min}$ along one path, a lower bound on L_J, not an estimate of it. With $\sigma_{\min}(J(\eta_0)) \geq 1.366$ (the smallest of the 10 tasks), if L_J equalled the largest secant a 2-step calibration error would move ident by at most $0.0237 \times 2/1.366 = 0.035$. Measured. Re-scoring all 70 camera units of the sweep with the Jacobian at $\hat{\eta}$ (vendor/score_selfmeasured.py, policy-free, the true-pose version recomputed in the same run) gives $|\text{ident}(\hat{\eta}) - \text{ident}(\eta)|$ median 0.000, maximum 0.159 on the 40 units calibrated within 2 cm (errors 0.01 to 1.10 cm), and median 0.087, maximum 0.811 on the 30 that fail, where the self-measured factor overstates identifiability (mean 0.91 against 0.70). The proposition is therefore used as a gate: the deployable score is $\text{ident}(\hat{\eta}) \times \hat{c} \times \mathbb{1}[\rho \leq \tau]$ with ρ the calibration's final silhouette residual (logged as cal_final_score; no simulator truth), τ the midpoint between the largest residual of a successful and the smallest of a failed OFT calibration (0.204 and 0.890), frozen before it was applied to either held-out family (Sec. 4.1; selfmeasured_gate.py).

Why the residual may gate: a self-calibration certificate†. The gate is not a heuristic threshold; in the local model it certifies the pose error, through the same singular value the identifiability factor uses.

Proposition 2 (residual certificate). Let $s(\eta) \in \mathbb{R}^m$ be the self-observation stacked over the sweep poses, C^1 with Jacobian $J(\eta)$ that is L-Lipschitz in operator norm on the segment $[\eta, \hat{\eta}]$; let the observation be $o = s(\eta) + \varepsilon$ and let $\hat{\eta}$ be any pose, with residual $r = s(\hat{\eta}) - o$ and $d = \hat{\eta} - \eta$. If $\|d\| \leq \sigma_{\min}(J(\eta))/L$, then

$$\|\hat{\eta} - \eta\| \leq \frac{2\,(\|r\| + \|\varepsilon\|)}{\sigma_{\min}(J(\eta))}. \tag{8}$$

Proof. By Taylor's theorem with a Lipschitz Jacobian, $s(\hat{\eta}) - s(\eta) = J(\eta)\,d + q$ with $\|q\| \leq \frac{L}{2}\|d\|^2$, so $\|s(\hat{\eta}) - s(\eta)\| \geq \sigma_{\min}(J(\eta))\|d\| - \frac{L}{2}\|d\|^2 \geq \frac{1}{2}\sigma_{\min}(J(\eta))\|d\|$ when $\|d\| \leq \sigma_{\min}(J(\eta))/L$. Since $s(\hat{\eta}) - s(\eta) = r + \varepsilon$, the triangle inequality gives $\frac{1}{2}\sigma_{\min}(J(\eta))\|d\| \leq \|r\| + \|\varepsilon\|$. □

With $\rho = \|r\|^2/m$ the logged mean-squared residual, a residual below τ certifies $\|d\| \leq 2(\sqrt{m\tau} + \|\varepsilon\|)/\sigma_{\min}$: the residual says the pose was measured, $\sigma_{\min}$ (the numerator of ident) says how tightly that certifies it, so the score's two factors are the two terms of one bound. The certificate is local: a pose in a wrong basin, farther than $\sigma_{\min}/L$, can fit with a small residual, which is why the refined pose source restarts from the training pose and keeps the lower loss (Sec. 4.4). Check (certificate_check.py): on the 89 camera units with a logged residual and a self-measured $\sigma_{\min}$, the certificate $\sqrt{\rho}/\sigma_{\min}$ ranks the calibration error at Spearman 0.899 and separates all 44 failures (AUROC 1.000); ρ alone gives 0.895 and 1.000, since $\sigma_{\min}$ varies little across these units, and $1/\sigma_{\min}$ alone 0.395.

invert per channel. Camera. With $\hat{\eta}_c$ the self-calibrated pose and D the deployed depth, every deployed pixel is back-projected, moved to the training camera and splatted over a 2×2 pixel footprint (nearest depth wins). With $F \subseteq \Omega$ the training-view pixels that receive at least one point,

$$\text{invert}_{\text{cam}} = \hat{c} = |F|/|\Omega|, \tag{9}$$

with Ω the 256×256 training-view pixel grid (the observation resolution) and robot pixels counted as covered like any other; c denotes the same quantity at the true pose. For the score, $\hat{c}$ is computed on one frame per unit, the first frame after the measurement sweep; the in-rollout coverage of the LIBERO-Plus camera rows (Fig. 2b) is the mean of (9) over the

episode's corrected frames. Single-frame coverage is a proxy for coverage over the states the task visits (the definition of invertibility), since later states are not measured. Light. Split the robot mask at the middle row of its extent into top and bottom halves; with per-channel means $\mu^{\mathrm{t}}, \mu^{\mathrm{b}}$ and standard deviations $s^{\mathrm{t}}, s^{\mathrm{b}}$ of the halves and s of the whole body (all in grey levels, $\epsilon = 10^{-3}$),

$$D(o) = \sum_{c=1}^{3} \Big| \log \frac{s_c^{\mathrm{t}} + \epsilon}{s_c^{\mathrm{b}} + \epsilon} \Big| + \sum_{c=1}^{3} \frac{|\mu_c^{\mathrm{t}} - \mu_c^{\mathrm{b}}|}{s_c + \epsilon}, \qquad \mathrm{invert}_{\mathrm{light}} = \frac{\min(D, D_0)}{\max(D, D_0)}, \tag{10}$$

natural logarithm, D_0 the same statistic in the training setup; $\mathrm{invert}_{\mathrm{light}}$ is undefined (the unit is dropped) when the mask occupies fewer than 4 rows, a half has fewer than 50 pixels, or D or D_0 is 0 or undefined. On real-valued, unclipped pixel values, D is exactly invariant to a per-channel map $x \mapsto g_c x + o_c$ with $g_c > 0$ when $\epsilon = 0$ (spreads scale by g_c in both halves, mean differences by g_c against a spread scaled by g_c); 8-bit quantisation makes it approximate, and clipping at 0 and 255 breaks it where pixels saturate. The per-channel affine map is the only change the photometric channel inverts. Noise. $\nu(o) = 1.4826 \cdot \mathrm{MAD}\big(r\big)$, $r = g - \mathrm{med}_{3\times3}(g)$, on the whole grey image g, in grey levels, with $\mathrm{MAD}(r) = \mathrm{med}\,|r - \mathrm{med}\, r|$; for Gaussian r, $1.4826\,\mathrm{MAD}(r)$ is a consistent estimate of the standard deviation of r (a proxy for the added noise level, not equal to it, since the median-filter residual is not the noise itself). With ν_0 the training floor and $\tilde{o}$ the denoised frame,

$$\mathrm{invert}_{\mathrm{noise}} = \begin{cases} 1 & \nu(o) \le 2\nu_0 + 1, \\ \mathrm{clip}_{[0,1]}\Big(1 - \dfrac{\max(\nu(\tilde{o}) - \nu_0,\, 0)}{\nu(o) - \nu_0}\Big) & \text{otherwise}, \end{cases} \tag{11}$$

the share of the added noise level that the denoiser removes. Joint offset: invert = 1 (the servo restores the encoders exactly). Scene content: invert = 0 (no inversion exists). The score is recover = ident × invert $\in [0, 1]$ and uses no policy output.

Static residual and calibration loss. With M_{obs} the observed robot mask at configuration q and $M(q, \eta)$ the mask the body model renders at camera pose η, let $\partial M = M \setminus \mathrm{erode}_{3\times3}(M)$ and $\mathrm{dt}_{\partial M}(p)$ the distance from pixel p to ∂M (Euclidean, 3×3 distance-transform approximation). Then

$$r(q; \eta) = \tfrac{1}{2}\Big(\operatorname*{mean}_{p \in \partial M_{\mathrm{obs}}} \mathrm{dt}_{\partial M(q,\eta)}(p) + \operatorname*{mean}_{p \in \partial M(q,\eta)} \mathrm{dt}_{\partial M_{\mathrm{obs}}}(p) \Big) + 4 \Big| \log \frac{|M(q,\eta)| + 1}{|M_{\mathrm{obs}}| + 1} \Big|, \tag{12}$$

in pixels at the mask resolution, natural logarithm; the chamfer is set to 10^3 if either boundary is empty, and under learned sensing the rendered mask passes through the same cleaning as the observed one. The first term is the symmetric mean boundary chamfer, the second penalises a change of apparent size (a camera that backs away shrinks the chamfer). The static residual† is $r(q) = r(q; \eta_0)$ at one rest pose; the motion-free variant skips the sweep iff $r \le \tau = 1.5$ (chosen post hoc from logged rows). The calibration loss over the K swept poses and its acceptance rule are

$$L(\eta) = \tfrac{1}{K} \textstyle\sum_k r(q_k; \eta), \qquad \text{apply } \hat{\eta}_c \iff L(\hat{\eta}_c) < 0.8\, L(\eta_0) \text{ and } L(\eta_0) - L(\hat{\eta}_c) > 0.5. \tag{13}$$

The pose-averaged score logged as cal_nominal_score is the calibration loss at the training pose,

$$S = L(\eta_0) = \tfrac{1}{K} \textstyle\sum_k r(q_k; \eta_0), \tag{14}$$

over the K poses of the row's measurement sweep ($K = 3$ for the evaluated wrapper); it is a different statistic from the one-pose static residual $r(q)$.

Self-check. With W the observed robot pixels warped into the training camera with $\hat{\eta}_c$ and depth, and P the robot's kinematic silhouette rendered at the training camera (inside the area the warp can fill), $\mathrm{F1} = 2|W \cap P|/(|W| + |P|)$. The extension full_verify re-measures with the 7-pose sweep when F1 < 0.75 (frozen 2026-09-19 from policy-free data) and keeps the calibration with the higher F1; it also settles the arm before handing over. On DROID the self-check statistic is the IoU between the robot rendered at the estimated pose and the observed mask.

Fingerprint distance and tolerance. At each of $K = 7$ probe end-effector positions (a cross of ± 10 cm in x and y, $+8$ and -6 cm in z around the nominal position) the policy returns an action chunk; its probe action $a_k \in \mathbb{R}^3$ is the mean of the translational components over the chunk. Stacking gives $A \in \mathbb{R}^{K\times 3}$, and against the reference A_0 (the same task and instruction in its training setup, raw frame)

$$d(A, A_0) = \frac{\|A - A_0\|_F}{\max(\|A_0\|_F,\, 10^{-9})}. \tag{15}$$

The frozen rule keeps the raw frame if $d_{\text{raw}} \le 0.10$, and otherwise the frame with the smaller d; a chosen frame with $d > 0.25$ is flagged as expected to fail. The fingerprint AUROC is (4) with score $-d$ and label wrapped success, over the probed (task, condition) units of a family (69 each in Table 7). The rollout-free tolerance probe† of Fig. 4 uses (15) with $K = 1$: the mean translational action of the first chunk at the episode's untouched start state, with the true-pose correction, against a different reference, the frame corrected with the simulator's true camera pose, same instruction and sampler seed. This is not the fingerprint's centre probe (Table 7, row 3), which is taken after the measurement sweep returns the arm to the training end-effector pose (17 to 25 mm from the start on some tasks), on the self-measured correction. No $K = 1$ null exists in the rows: the reference queried twice with the same seed gives $d = 0$ exactly on all 40 runs. The knee, the dose at which the median opening action changes by 10%, is $\delta^\star = \min\{\delta \text{ on the grid} : \text{median}_{\text{units}}\, d(\delta) > 0.10\}$, the median over units of each unit's median over the three axes. The threshold 0.10 was frozen for $K = 7$; for $K = 1$ it is a reference line, and the knee moves with it (results/drive_vihan/tolerance/tolerance_*.jsonl, 20 units per family): at $\tau = 0.05, 0.10, 0.20$ the knee is 0.5, 1 and 4 cm and 0.5, 1 and 4° for OFT, and 0.5, 2 and 4 cm and 0.5, 0.5 and 4° for $\pi_{0.5}$. Lemma 6 says what the knee is a statistic of. Coordinates of the probe field. Probe positions p_k are end-effector positions in metres (world frame). Probe actions are the policies' outputs after their own un-normalisation, i.e. LIBERO controller commands (OSC delta, position part in $[-1, 1]$, scaled by the controller's output limit), not the q_{01}/q_{99}-normalised outputs of the networks; the chunk has 8 actions for OFT and 1 for $\pi_{0.5}$ (LeRobot's select_action returns one action of its queue per call) and for OpenVLA. $\hat{A}$ of Prop. 6 therefore has units of command per metre; a positive scalar change of units leaves every sign unchanged. Calibration error is the position error $\|\hat{t} - t\|_2$ of the camera centre in cm, on rows where the pose was applied (cal_applied); "within 2 cm" is the share of those rows with error ≤ 2 cm.

Other measured quantities†. Shift magnitude of a camera condition with true pose (t, R) and training pose (t_0, R_0): $s_{\text{trans}} = \|t - t_0\|_2$ (cm) and $s_{\text{rot}} = \angle(\mathsf{R}\mathsf{R}_0^\top)$ (degrees, the rotation angle), scored as $-s$ for the baselines of Sec. 4.1. Triangulation residual: with p_k the probe positions, u_k the unit commanded directions of the probes whose command does not vanish and $\hat{x}$ the least-squares meeting point of Prop. 6, $\text{res} = (\frac{1}{K}\sum_k \angle(u_k, \hat{x} - p_k)^2)^{1/2}$ in degrees (smc.triangulate). The centre probe is p_1, the nominal end-effector position, throughout. Image error of a frame $\hat{o}$ against a reference render o_{ref}: $\frac{1}{|\Omega|}\sum_{u\in\Omega}\frac{1}{3}\sum_{c=1}^{3}|\hat{o}_c(u) - o_{\text{ref},c}(u)|$ on 0 to 255 (vendor/fidelity.py; the same mean restricted to a pixel set for the seen and hole errors). Placements: the envelope renders 88 camera placements (64 on an azimuth-by-distance grid, 16 elevations, 8 look-aways) on 3 base tasks, 264 task-placements; the 192 of Appendix D are the 64 grid placements on the 3 tasks, excluding elevations and look-aways. Selector success over U units is $|U|^{-1}\sum_u Y_u$(chosen frame), and the per-unit oracle is $|U|^{-1}\sum_u \max(Y_u^{\text{r}}, Y_u^{\text{w}})$.

Tests and intervals. Paired AUROC difference: $\Delta = \widehat{A}(s) - \widehat{A}(s')$ (Eq. 4) on the units where both scores are defined; its interval is the 2.5 and 97.5 percentiles of Δ recomputed on the same bootstrap resample for both scores, resampling units or, for the task-cluster version, base tasks with all their units. Task-level tests for replicated rollouts: with $D_i = \sum_\ell (Y_{i\ell}^{\text{w}} - Y_{i\ell}^{\text{r}})$ the summed paired difference of task i over replicates (or over its units), keep the n_0 tasks with $D_i \neq 0$, n_+ positive and n_- negative. The sign test is (3) with $b = n_+$, $c = n_-$; the exact sign-flip test is

$$p = 2^{-n_0}\,\#\Big\{s \in \{\pm 1\}^{n_0} : \Big|\sum_i s_i D_i\Big| \ge \Big|\sum_i D_i\Big|\Big\}, \tag{16}$$

and the Wilcoxon signed-rank test uses $W = \sum_{D_i>0} \mathrm{rank}(|D_i|)$ (mid-ranks) with the exact null over the same 2^{n_0} sign assignments, two-sided. The null has 2^{n_0} equally likely points, so the smallest attainable p is 2^{1-n_0}. The sign test ignores magnitudes, the sign-flip test weights each task by $|D_i|$, and Wilcoxon uses their ranks, so on the same D_i they can differ widely (App. D). Wilson interval for k of n: $\big(\hat{p}+\frac{z^2}{2n} \pm z\sqrt{\hat{p}(1-\hat{p})/n + z^2/(4n^2)}\big)/(1+z^2/n)$, $\hat{p} = k/n$, $z = 1.96$. One-sided Clopper-Pearson upper limit for k events in n: the u with $\Pr(\mathrm{Bin}(n,u) \le k) = \alpha$; for $k = 0$ it is $1 - \alpha^{1/n}$ (0 of 20 gives 0.139). Holm: with the $m = 15$ secondary p values of Table 6 sorted, $p_{(1)} \le \cdots \le p_{(m)}$, the adjusted value is $p^{\mathrm{H}}_{(i)} = \max_{j\le i} \min\big(1, (m-j+1)\,p_{(j)}\big)$. Log-log slope: the OLS slope $\hat{b} = \sum_r (x_r - \bar{x})(y_r - \bar{y}) / \sum_r (x_r - \bar{x})^2$ with $x = \ln \sigma_{\min}$ and $y = \ln$(calibration error in cm) over runs r, t-interval with $n-2$ degrees of freedom, and a task-cluster bootstrap (4,000 resamples of the 8 tasks). Equal-mass bins (Fig. 2b): the $n = 98$ units sorted by $\hat{c}$ are split into five consecutive groups of 20, 20, 20, 19 and 19, each plotted at its mean $\hat{c}$ against its repaired share. Flip rate: $\hat{f}$ = (units with nothing applied whose raw and wrapped outcomes differ) / (units with nothing applied) on the OFT graded sweep, $7/53 = 0.132$, one-sided 95% limit 0.234; the same count on $\pi_{0.5}$ is 15 of 48.

Attribution ladder†. Rungs $k = 0, 1, 2, 3$ are raw, the correction with the measured pose, the same reprojection with the simulator's true pose (variant cam_true_ref: training-view fill, the fill the measured rung uses; the variant with the deployed frame's own fill scores 15 of 30 on set A), and the training camera's own render. With $X_k(u) \in \{0, 1\}$ the outcome of rung k on a raw failure u (so $X_0(u) = 0$),

$$\beta(u) = \min\{k \in \{1, 2, 3\} : X_k(u) = 1\}, \qquad \beta(u) = 4 \text{ (“policy”) if no such } k. \tag{17}$$

The buckets $\beta = 1, 2, 3, 4$ are read “first repaired at the measured pose”, “not identified”, “not invertible” and “policy”. The ladder is monotone on u if $X_1(u) \le X_2(u) \le X_3(u)$.

B.2 The four facts of Section 3.1: statements and proofs

Section 3.1 states these results in one sentence each; the statements, with their conditions, are the following. $z(q, \eta)$ is the self-observation at joint configuration q (Appendix B.1), the robot chooses $q_1, \ldots, q_K$, and J is the Jacobian of the stacked z in the nuisance parameters.

Proposition 3 (measurement error from the body alone). Let $z = z_0 + J\delta + \epsilon$ with J the stacked Jacobian over $q_1, \ldots, q_K$ of full column rank d_η (the number of nuisance parameters) and $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$. Least squares is unbiased with $\mathbb{E}\|\hat{\delta} - \delta\|^2 = \sigma^2 \,\mathrm{tr}\big((J^\top J)^{-1}\big)$, so $\sigma^2/\sigma_{\min}(J)^2 \le \mathbb{E}\|\hat{\delta} - \delta\|^2 \le d_\eta\, \sigma^2/\sigma_{\min}(J)^2$: the root mean squared error is $\Theta(1/\sigma_{\min})$ within a factor $\sqrt{d_\eta}$ ($d_\eta = 6$ for a camera). Adding a pose never lowers $\sigma_{\min}(J)$.

Proposition 8 below restates it for whitened observations $\Sigma^{-1/2} z$, which is the form that matches the blocks of z; the calibration minimises the chamfer loss L of Eq. 13, so the Jacobian that governs its error is $D_\eta r(q_k; \eta)$ stacked over k, and both propositions are local models of it.

Proposition 4 (what the deployed frame can contribute). Let $(s, o_d, o_{\mathrm{train}}, R)$ be jointly distributed (s the state, o_d the deployed frame, o_{train} the training-view frame), pixel values in $[0, 255]$ per channel, f measurable, and let $V \subseteq \Omega$ be a set of training-view pixels (we take $V = F$, the splat set of Eq. 9 at the true pose; Lemma 1), $c = |V|/|\Omega|$, and R any stored side information. Any restoration $\hat{o} = f(o_d, R)$ of the training view from the deployed frame o_d satisfies $\frac{1}{|\Omega|}\mathbb{E}\sum_u |\hat{o}(u) - o_{\mathrm{train}}(u)| \ge (1 - c)\,\bar{e}^\star$, with $\bar{e}^\star$ the mean over $\Omega \setminus V$ of the conditional Bayes L_1 risk given (o_d, R). If that risk is at least $e_{\min}$ on every pixel outside V, at most a fraction c of the training view can be restored to expected error below $e_{\min}$.

Proposition 4 bounds what the deployed frame alone can contribute; the operative statement is the exact decomposition of Prop. 12 below, of which the bound is a remark. Fig. 5 shows that the training-view fill makes the bound slack (holes cost 7.6 on 0 to 255), and the relation between coverage and repair (Fig. 2b) is empirical.

Proposition 5 (a motion-free test for "the camera moved"). Assume the body model is exact and the sensing and encoder errors are bounded, so that $\nu_{\max} = \sup r(q;\eta_0) < \infty$ (supremum over error realisations with the camera at η_0; see Prop. 18 for unbounded noise). If the camera is at η_0, the silhouette residual $r(q)$ (Eq. 12) between the observed mask and the mask rendered at η_0 at the encoder reading q is a function of sensing and encoder error alone. A residual that exceeds $\nu_{\max}$ implies $\eta \neq \eta_0$; with an estimated floor this is a test whose error rates Proposition 18 bounds. One pose does not identify η, but it decides whether identification is needed.

Proposition 6 (a diverging opening field has no sub-goal). Let the opening action near the probes be $a(p) = a_0 + A(p - p_1)$, p_1 the centre probe, and $g^\star = p_1 - A^{-1}a_0$. (i) Given the probe positions, the triangulation point and residual are functions of the fitted field (a_0, A) only, hence cannot register miscalibration except through the field; if $A = \mu I$, every probe ray passes through $g^\star$, which lies behind every probe iff $\mu > 0$, and the residual is zero at any camera. (ii) If $A = \mu I + E$ with $\|E\| \le \varepsilon|\mu|$, the least-squares point is within $K\varepsilon r_{\max}/\lambda_{\min}(M)$ of $g^\star$ and the residual vanishes with E. (iii) If $\operatorname{tr} A > 0$, $g^\star$ is not an attractor of $\dot p = a(p)$, and for the discrete update $p_{t+1} = p_t + \alpha\, a(p_t)$ an eigenvalue λ of A with $\operatorname{Re}\lambda > 0$ gives $|1 + \alpha\lambda| > 1$ for every $\alpha > 0$, so $g^\star$ is not asymptotically stable for any step size.

The proofs follow; Proposition 14 adds the invariances of (i) and the discrete step of (iii).

Proposition 3. Assume $z(\eta_0 + \delta) = z_0 + J\delta + o(\|\delta\|)$ (smooth self-observations, local linearisation), $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$ i.i.d., and $J \in \mathbb{R}^{m \times d_\eta}$ of full column rank. Dropping the remainder, $\hat\delta = (J^\top J)^{-1} J^\top (z - z_0) = \delta + (J^\top J)^{-1} J^\top \epsilon$, so $\mathbb{E}\hat\delta = \delta$ and $\operatorname{Cov}(\hat\delta) = \sigma^2 (J^\top J)^{-1}$, which is also the Cramér-Rao bound for any unbiased estimator. Hence $\mathbb{E}\|\hat\delta - \delta\|^2 = \sigma^2 \sum_{i=1}^{d_\eta} \lambda_i^{-1}$ with λ_i the eigenvalues of $J^\top J$; the largest term is $\lambda_{\min}^{-1} = \sigma_{\min}(J)^{-2}$ and there are d_η terms, which gives both bounds. Appending the rows of a new pose gives $J'^\top J' = J^\top J + B$ with $B = J_{K+1}^\top J_{K+1} \succeq 0$, and by Courant-Fischer $\lambda_{\min}(J^\top J + B) = \min_{\|x\|=1} x^\top (J^\top J + B) x \ge \min_{\|x\|=1} x^\top J^\top J x = \lambda_{\min}(J^\top J)$. The calibration we run is a biased search (coarse grid, cross-entropy, ICP); for it the matching statement is the Bayesian bound of Trees (1968) with the search's prior, which has the same $(J^\top J)^{-1}$ dependence plus the prior information. The testable consequence is error $\approx C/\sigma_{\min}$, a log-log slope of -1 (rows in results/local/measurability.jsonl, 8 tasks, $K \in \{1, 2, 3, 5, 7\}$, probe poses spanning the full and the half reach used by the method, 40 and 32 distinct runs). Check. On the 8 camera tasks, adding poses raises the median $\sigma_{\min}$ monotonically (1.04, 1.71, 2.38, 3.11, 3.61 at $K = 1, 2, 3, 5, 7$) while the median calibration error falls non-monotonically (0.54, 0.17, 0.28, 0.11, 0.13 cm; the method uses $K = 3$). The OLS slope (App. B.1) is -0.84, t-interval $[-1.60, -0.08]$, task-cluster bootstrap $[-1.58, -0.22]$, at full reach, and -0.19 $[-1.14, 0.75]$ (cluster $[-1.07, 0.49]$) at half reach: consistent with -1 at full reach but weakly determined, and a change of spectrum shape across K could produce the same slope (Prop. 7). Pooled over the 72 distinct runs (the 8 half-reach $K = 1$ runs repeat the full-reach ones exactly: the one probe is the rest pose, so the reach does not enter), Spearman $\rho_s(\sigma_{\min}, \text{error}) = -0.21$; the runs share tasks, so instead of the naive p we permute errors within tasks (4,000 permutations), $p = 0.006$ (measurability_dedupe.py; 80 rows with the duplicates gave -0.25, $p = 0.001$). The proposition's quantity is $\operatorname{tr}((J^\top J)^{-1})$, which the logged $\sigma_{\min}$ and $\sigma_{\max}$ bracket (Prop. 8, (c)); regressing ln(error) on $\frac{1}{2}$ ln of the lower and of the upper bracket gives slopes 0.85 and 0.84 (both [0.09, 1.60]) at full reach, against the predicted $+1$, and 0.20 and 0.19 (both $[-0.75, 1.14]$) at half reach. Per K, the product of the medians $\sigma_{\min} \times$ error is 0.56, 0.30, 0.66, 0.33, 0.46 at full reach; at half reach $\sigma_{\min}$ rises 1.04 to 3.29 and the error goes 0.54, 0.25, 0.47, 0.41, 0.21 cm. The worst task at $K = 7$ (full reach) is 1.2 cm. The departures from monotonicity at $K = 3$ are the search, not the information: the bound is on the variance of an efficient estimator.

The slope test of Prop. 3†.

Proposition 7 (expected log error). Under the model of Prop. 8 with $\Sigma = \sigma^2 I$, write the eigenvalues of $\tilde J^\top \tilde J$ as $\sigma_1^2 \ge \cdots \ge \sigma_{d_\eta}^2 = \sigma_{\min}^2$ and $s_i = \sigma_{\min}^2/\sigma_i^2 \in (0, 1]$ (so $s_{d_\eta} = 1$). Then,

natural logarithm,

$$\mathbb{E}\ln\|\hat{\delta}-\delta\| = -\ln\sigma_{\min} + g(s), \qquad g(s) = \tfrac{1}{2}\,\mathbb{E}\ln\Big(\sum_i s_i\,\xi_i^2\Big), \quad \xi\sim\mathcal{N}(0, I_{d_\eta}), \tag{18}$$

g is nondecreasing in each s_i, and $C \le g(s) \le C + \Delta_{d_\eta}$ with $C = \frac{1}{2}(\psi(\frac{1}{2}) + \ln 2) = -\frac{1}{2}(\gamma + \ln 2) = -0.635$ and $C + \Delta_{d_\eta} = \frac{1}{2}(\psi(\frac{d_\eta}{2}) + \ln 2)$ ($= 0.808$ for $d_\eta = 6$, $\Delta_6 = 1.443$), ψ the digamma function and γ Euler's constant. Both ends are sharp: the upper at $s = \mathbf{1}$ (isotropic), the lower in the limit $s_i \to 0$ for $i < d_\eta$. The camera-centre error $\|\hat{\delta}_t - \delta_t\|$ (the three translation coordinates, the logged quantity) satisfies $\mathbb{E}\ln\|\hat{\delta}_t - \delta_t\| \le -\ln\sigma_{\min} + g(s)$ and, with $h_3 = \frac{1}{2}(\psi(\frac{3}{2}) + \ln 2) = 0.365$,

$$-\ln\sigma_{\min} + \tfrac{1}{2}\ln s_1 + h_3 \;\le\; \mathbb{E}\ln\|\hat{\delta}_t - \delta_t\| \;\le\; -\ln\sigma_{\min} + h_3, \qquad s_1 = \sigma_{\min}^2/\sigma_{\max}^2. \tag{19}$$

Proof. $\hat{\delta} - \delta \sim \mathcal{N}(0, (\tilde{J}^\top\tilde{J})^{-1})$; in the eigenbasis of $\tilde{J}^\top\tilde{J}$, $\|\hat{\delta} - \delta\|^2 = \sum_i \sigma_i^{-2}\xi_i^2 = \sigma_{\min}^{-2}\sum_i s_i\xi_i^2$, and half the log of this gives (18). Monotonicity is that of ln. Since $s_{d_\eta} = 1$ and $s_i \le 1$, $\xi_{d_\eta}^2 \le \sum_i s_i\xi_i^2 \le \sum_i \xi_i^2$; $\mathbb{E}\ln\chi_k^2 = \psi(k/2) + \ln 2$ gives the two ends, attained at the stated spectra (dominated convergence for the lower one; $\mathbb{E}|\ln\xi_{d_\eta}^2| < \infty$). The translation error is a coordinate projection, so its norm is at most $\|\hat{\delta} - \delta\|$. For the second range: $\hat{\delta}_t - \delta_t \sim \mathcal{N}(0, B)$ with B the 3×3 principal block of $(\tilde{J}^\top\tilde{J})^{-1}$, whose eigenvalues lie in $[\sigma_{\max}^{-2}, \sigma_{\min}^{-2}]$ by Cauchy interlacing; so $\|\hat{\delta}_t - \delta_t\|^2 = \sigma_{\min}^{-2}\sum_{j\le 3} t_j\zeta_j^2$ with $\zeta\sim\mathcal{N}(0, I_3)$ and $t_j \in [s_1, 1]$, and monotonicity with $\mathbb{E}\ln\chi_3^2 = \psi(\frac{3}{2}) + \ln 2$ gives (19). □

So the slope of ln(error) on $\ln\sigma_{\min}$ across K is -1 only if the spectrum shape s is the same at every K; otherwise it is $-1 + \Delta g/\Delta\ln\sigma_{\min}$ (full pose error) or $-1 + \Delta h/\Delta\ln\sigma_{\min}$ with h the offset in (19) (camera-centre error). The form $C + \frac{1}{2}\ln d_\eta$ (0.26 for $d_\eta = 6$) is not an upper bound on g, since the isotropic value 0.808 exceeds it; Jensen's inequality gives the valid but looser $g \le \frac{1}{2}\ln\sum_i s_i \le \frac{1}{2}\ln d_\eta = 0.896$, and the width Δ_{d_η} is the sharp one. Check (results/local/measurability.jsonl, full reach, 8 tasks per K; the regressed error is err_cm, the camera-centre error). The logged $\sigma_{\min}/\sigma_{\max}$ fixes s_1 (median $s_1 = 0.015$ to 0.022 over $K = 1,\dots,7$) and leaves $s_2,\dots,s_{d_\eta-1} \in [s_1, 1]$, so $g \in [-0.34, 0.70]$ at every K (Monte Carlo, 2×10^5 draws), and the camera-centre offset $h \in [\frac{1}{2}\ln s_1 + 0.365, 0.365]$, i.e. $[-1.74, 0.365]$ at $K = 1$ and $[-1.55, 0.365]$ at $K = 7$. Between $K = 1$ and $K = 7$ the median $\ln\sigma_{\min}$ moves by $\ln(3.61/1.04) = 1.24$. For the full pose error g can move by up to $0.70 + 0.34 = 1.04$, a slope shift of up to ± 0.84; for the logged camera-centre error h can move by up to $0.365 + 1.74 = 2.10$, a slope shift of up to ± 1.7. The measured -0.84 [-1.60, -0.08] is consistent with -1, and also with a shape change, which the rows cannot exclude; the slope test constrains the spectrum shape only if the shape is logged.

Measurement error under whitened observations (restates Prop. 3)†. The blocks of z (mask, colour, joint, noise; Appendix B.1) have different units and noise levels, so the isotropic noise of Prop. 3 is a simplification. Let $\Sigma = \mathrm{blockdiag}(\Sigma_{\mathrm{m}}, \Sigma_{\mathrm{c}}, \Sigma_{\mathrm{q}}, \Sigma_\nu) \succ 0$ be the sensing covariance, the same at every pose, with $s_{\min}^2 = \lambda_{\min}(\Sigma)$, $s_{\max}^2 = \lambda_{\max}(\Sigma)$ (so $s_{\max}$ is the largest sensing-noise standard deviation along any direction of z).

Proposition 8 (whitened measurement error). Let $z = z_0 + J\delta + \epsilon$, $\epsilon\sim\mathcal{N}(0,\Sigma)$, $J\in\mathbb{R}^{m\times d_\eta}$ of full column rank, and $\tilde{J} = \Sigma^{-1/2}J$. The generalised least-squares estimate is unbiased with

$$\mathbb{E}\|\hat{\delta}-\delta\|^2 = \mathrm{tr}\big((\tilde{J}^\top\tilde{J})^{-1}\big), \qquad \frac{1}{\sigma_{\min}(\tilde{J})^2} \le \mathrm{tr}\big((\tilde{J}^\top\tilde{J})^{-1}\big) \le \frac{d_\eta}{\sigma_{\min}(\tilde{J})^2},$$
$$\frac{\sigma_{\min}(J)}{s_{\max}} \le \sigma_{\min}(\tilde{J}) \le \frac{\sigma_{\min}(J)}{s_{\min}}. \tag{20}$$

Consequently, if the nuisance leaves the sensing noise unchanged, $\Sigma(\eta) = \Sigma(\eta_0)$ (the camera and joint-offset families), then with ident_Σ the ratio (6) computed from $\tilde{J}$ and ident the one computed from the unwhitened step-scaled J (as implemented), $\mathrm{ident}/\kappa \le \mathrm{ident}_\Sigma \le \kappa\,\mathrm{ident}$ whenever both ratios are below 1, with $\kappa = s_{\max}/s_{\min}$. For light and noise, which change Σ, the generalised form is Prop. 9.

Proof. Whitening gives $\Sigma^{-1/2}(z - z_0) = \tilde{J}\delta + \tilde{\epsilon}$ with $\tilde{\epsilon} \sim \mathcal{N}(0, I)$, so Prop. 3 with $\sigma = 1$ and $\tilde{J}$ in place of J gives the first two statements (the GLS estimate is the OLS estimate of the whitened model). For the third, $\|\tilde{J}x\| = \|\Sigma^{-1/2}Jx\|$ lies between $\|Jx\|/s_{\max}$ and $\|Jx\|/s_{\min}$ for every x; take the minimum over $\|x\| = 1$. For the ratio, the upper bound on $\sigma_{\min}(\tilde{J}(\eta))$ divided by the lower bound on $\sigma_{\min}(\tilde{J}(\eta_0))$ gives $\text{ident}_\Sigma \leq \kappa\,\text{ident}$, and symmetrically $\text{ident}_\Sigma \geq \text{ident}/\kappa$. □

Three qualifications fix what the proposition covers. (a) ident uses the unwhitened, step-scaled $J_\phi = (\partial z/\partial\theta_\phi)\,W$, $W = \text{diag}(h)$; it is a proxy for $\sigma_{\min}(\tilde{J})$ exact only up to κ, and $\sigma_{\min}(J_\phi)$ is unit-dependent: replacing the steps h (1 cm, 1°) by Dh with D positive diagonal and not a multiple of I changes $\sigma_{\min}(J_\phi)$ and, in general, the ratio (6), while a common scalar cancels in the ratio. (b) The calibration does not solve least squares on z: it minimises the chamfer loss L of Eq. 13 by a biased search, so the Jacobian that governs its error is $D_\eta r(q_k;\eta)$ stacked over k, and Prop. 8 is a local model of it (the squared-error, Gaussian case of an M-estimator whose covariance is in general the sandwich $H^{-1}\,\text{Var}(\nabla L)\,H^{-1}$ with H the Hessian of L). (c) The quantity the proposition predicts is $\text{tr}((J^\top J)^{-1})$, not $\sigma_{\min}^{-2}$; the measurability rows log only $\sigma_{\min}$ and $\sigma_{\max}$, which bracket it, for $d_\eta \geq 2$, as $\sigma_{\min}^{-2} + \sigma_{\max}^{-2} \leq \text{tr}((J^\top J)^{-1}) \leq (d_\eta - 1)\sigma_{\min}^{-2} + \sigma_{\max}^{-2}$ (the extreme eigenvalues of $J^\top J$ are $\sigma_{\min}^2$ and $\sigma_{\max}^2$; the other $d_\eta - 2$ lie between).

When the nuisance changes the sensing noise†. Prop. 8's sandwich assumes $\Sigma(\eta) = \Sigma(\eta_0)$: the nuisance moves the mean of z, not its noise. This is the camera and joint-offset case (a moved camera or arm changes where the silhouette falls, not the sensor). Light and noise conditions change the sensing noise itself.

Proposition 9 (nuisance-dependent covariance). Let $\Sigma(\eta) \succ 0$ with extreme eigenvalues $s_{\min}(\eta)^2$, $s_{\max}(\eta)^2$, $\tilde{J}(\eta) = \Sigma(\eta)^{-1/2}J(\eta)$, and ident_Σ, ident the ratios (6) from $\tilde{J}$ and from J, both below 1. Then

$$\frac{s_{\min}(\eta_0)}{s_{\max}(\eta)}\,\text{ident} \;\leq\; \text{ident}_\Sigma \;\leq\; \frac{s_{\max}(\eta_0)}{s_{\min}(\eta)}\,\text{ident},$$
$$\text{so}\quad \text{ident}/\kappa' \leq \text{ident}_\Sigma \leq \kappa'\,\text{ident}, \qquad \kappa' = \max\Big(\frac{s_{\max}(\eta)}{s_{\min}(\eta_0)}, \frac{s_{\max}(\eta_0)}{s_{\min}(\eta)}\Big), \tag{21}$$

which reduces to Prop. 8 with $\kappa = s_{\max}/s_{\min}$ when $\Sigma(\eta) = \Sigma(\eta_0)$.

Proof. As in Prop. 8, $\sigma_{\min}(J(\eta))/s_{\max}(\eta) \leq \sigma_{\min}(\tilde{J}(\eta)) \leq \sigma_{\min}(J(\eta))/s_{\min}(\eta)$, and the same at η_0. The upper bound on the numerator of ident_Σ over the lower bound on its denominator gives $\text{ident}_\Sigma \leq \text{ident}\, s_{\max}(\eta_0)/s_{\min}(\eta)$; the lower bound on the numerator over the upper bound on the denominator gives $\text{ident}_\Sigma \geq \text{ident}\, s_{\min}(\eta_0)/s_{\max}(\eta)$. Both factors are at most κ'. □

If added noise inflates every block's standard deviation by a factor $g \geq 1$ ($\Sigma(\eta) = g^2\Sigma(\eta_0)$), then $\sigma_{\min}(\Sigma(\eta)^{-1/2}J(\eta)) = \sigma_{\min}(\Sigma(\eta_0)^{-1/2}J(\eta))/g$, so $\text{ident}_\Sigma = \text{ident}_{\Sigma_0}/g$ exactly, with ident_{Σ_0} the ratio whitened by $\Sigma(\eta_0)$ at both ends (both ratios below 1); this equals ident/g when $\Sigma(\eta_0)$ is isotropic, and otherwise lies in $[\text{ident}/(g\kappa), \kappa\,\text{ident}/g]$ by Prop. 8. The whitened score falls with the noise even when the unwhitened one does not. Check. Not computable: Σ is not logged. In the implementation z is read from noiseless renders except for its noise-level entry, which uses a fixed generator seed, so Σ here models the sensing of a real deployment, not a quantity of the simulator rows.

The secant the implementation computes†. The columns $J_{\text{fd}}(\eta)_{:,i} = z(\eta + h_i e_i) - z(\eta)$ are secants, not $(\partial z/\partial\theta_i)h_i$. Two readings make Props. 8 and 11 exact statements about what is computed. (a) Secant model, the one we adopt: define the local model on the step lattice, $z(\eta + Wu) = z(\eta) + J_{\text{fd}}u + \epsilon$ for u in step units; it is exact at $u = 0$ and $u = e_i$ and is the linear interpolation between them, and Props. 8 and 11 hold verbatim with $J = J_{\text{fd}}$ for this model. (b) Jacobian model, with the discrepancy carried as an additive term:

Proposition 10 (secant error through Weyl). Let z be twice continuously differentiable along each coordinate segment, $M_i = \sup_{0\le t\le h_i} \|\partial_i^2 z(\eta + te_i)\|$, and $\varepsilon_{\mathrm{fd}} = \frac{1}{2}\big(\sum_i h_i^4 M_i^2\big)^{1/2}$. Then $\|J_{\mathrm{fd}} - JW\|_2 \le \varepsilon_{\mathrm{fd}} \le \frac{\sqrt{d_\eta}}{2} \max_i h_i^2 M_i$, $|\sigma_{\min}(J_{\mathrm{fd}}) - \sigma_{\min}(JW)| \le \varepsilon_{\mathrm{fd}}$, and under the model of Prop. 8 with the Jacobian JW,

$$\Pr\big(\|\hat{u} - u\| \le t\big) \;\ge\; 1 - \frac{d_\eta\, s_{\max}^2}{\big(\mathrm{ident}\,\sigma_{\min}(J_{\mathrm{fd}}(\eta_0)) - \varepsilon_{\mathrm{fd}}(\eta)\big)_+^2\, t^2}, \tag{22}$$

with ident as implemented (from the secants) and the right-hand side read as $-\infty$ when the bracket is 0.

Proof. Taylor's theorem, applied to each coordinate of z along e_i and bounded in norm by the integral form of the remainder, gives $\|z(\eta + h_i e_i) - z(\eta) - h_i \partial_i z(\eta)\| \le \frac{1}{2} h_i^2 M_i$, so the i-th column of $E = J_{\mathrm{fd}} - JW$ has norm at most $\frac{1}{2} h_i^2 M_i$, and $\|E\|_2 \le \|E\|_F \le \varepsilon_{\mathrm{fd}} \le \frac{\sqrt{d_\eta}}{2} \max_i h_i^2 M_i$. Weyl's inequality gives the singular-value bound. By (6) applied to the secants, $\sigma_{\min}(J_{\mathrm{fd}}(\eta)) \ge \mathrm{ident}\,\sigma_{\min}(J_{\mathrm{fd}}(\eta_0))$, so $\sigma_{\min}(JW) \ge \mathrm{ident}\,\sigma_{\min}(J_{\mathrm{fd}}(\eta_0)) - \varepsilon_{\mathrm{fd}}(\eta)$; insert this in the proof of Prop. 11 ($\mathrm{tr}((\tilde{J}^\top \tilde{J})^{-1}) \le d_\eta\, s_{\max}^2/\sigma_{\min}(JW)^2$) and apply Markov's inequality. □

Check. Second differences are not logged, so $\varepsilon_{\mathrm{fd}}$ is not available; the bound (22) stays non-vacuous for t large enough whenever $\varepsilon_{\mathrm{fd}} < \mathrm{ident} \times \sigma_{\min}(J_{\mathrm{fd}}(\eta_0))$, and the reference values of the camera block on the 10 sweep tasks are at least 1.366, median 1.851 (scores_v2.jsonl, nominal rows). Reading (a) needs no such condition: Props. 8 and 11 are statements about the secant model.

Pose error from ident: a Markov bound (on the squared error)[†].

Proposition 11 ($\Pr(E_1)$ as a function of ident). Under the model of Prop. 8 with $\hat{\delta}$ the GLS estimate and $J_\phi(\eta_0)$ the reference Jacobian (at η_0, same block, same steps), for every $t > 0$

$$\Pr\big(\|\hat{\delta} - \delta\| \le t\big) \;\ge\; 1 - \frac{\mathrm{tr}((\tilde{J}^\top \tilde{J})^{-1})}{t^2} \;\ge\; 1 - \frac{d_\eta\, s_{\max}^2}{\mathrm{ident}^2\, \sigma_{\min}(J_\phi(\eta_0))^2\, t^2}, \tag{23}$$

with ident as implemented (unwhitened). The right-hand side is increasing in ident and vacuous for ident $\le \sqrt{d_\eta}\, s_{\max}/(\sigma_{\min}(J_\phi(\eta_0))\, t)$. With $\sigma^2 I$ in place of Σ ($s_{\max} = \sigma$) it reads $1 - d_\eta \sigma^2/(\mathrm{ident}^2 \sigma_{\min}(J_\phi(\eta_0))^2 t^2)$.

Proof. Markov's inequality on $\|\hat{\delta} - \delta\|^2$ and Prop. 8 give the first inequality. By definition (6), $\sigma_{\min}(J) \ge \mathrm{ident}\,\sigma_{\min}(J_\phi(\eta_0))$: if the ratio is below 1 this is equality, otherwise ident = 1 and $\sigma_{\min}(J) \ge \sigma_{\min}(J_\phi(\eta_0))$. Then $\mathrm{tr}((\tilde{J}^\top \tilde{J})^{-1}) \le d_\eta/\sigma_{\min}(\tilde{J})^2 \le d_\eta\, s_{\max}^2/\sigma_{\min}(J)^2 \le d_\eta\, s_{\max}^2/(\mathrm{ident}^2 \sigma_{\min}(J_\phi(\eta_0))^2)$ by (20). □

The event E_1 of "Why a product" (translation error below the policy's tolerance, 2 cm here) contains $\{\|\hat{\delta} - \delta\| \le t\}$ for $t = 2$ step units, since the translation error is a coordinate projection of $\hat{\delta} - \delta$; (23) is therefore a lower bound on $\Pr(E_1)$ that increases with ident. It is a bound: it does not make $\Pr(E_1)$ itself monotone in ident, and it is stated for an efficient estimator, not for the search we run. Check (in-rollout sweep rows with cal_applied and a logged cam_est_err_cm_deg, ident from results/drive_vihan/sweep/scores_v2.jsonl; repeats of a (task, condition) averaged by median). OFT has 21 such units (15 further applied rows do not log the error; the 55 recovered rows log neither), and 18 of the 21 have ident = 1, so terciles are not defined without breaking ties arbitrarily; we bin by ident < 1 split at its median, and ident = 1. Share with error ≤ 2 cm (Wilson 95%): OFT 0/2 [0.00, 0.66], 0/1 [0.00, 0.79], 9/18 [0.29, 0.71] (all 3 units with ident < 1 are look_away_55); OpenVLA 0/3 [0.00, 0.56], 1/2 [0.09, 0.91], 19/23 [0.63, 0.93]; $\pi_{0.5}$ 3/20 [0.05, 0.36], 15/19 [0.57, 0.91], 16/31 [0.35, 0.68]. The shares are nondecreasing on OFT and OpenVLA, on 3 and 5 units below ident = 1, and are not nondecreasing on $\pi_{0.5}$, the only family with enough units below

1 (39): its top bin (ident = 1) does worse than the middle. Spearman(ident, error) is -0.58 ($p = 0.006$), -0.58 ($p = 0.001$) and -0.20 ($p = 0.09$) on OFT, OpenVLA and $\pi_{0.5}$. The rows support the direction of (23) and not a monotone $\Pr(E_1)$; a unit with ident = 1 still misses 2 cm about half the time on OFT (9 of 18) and $\pi_{0.5}$ (15 of 31), which the bound allows, since it only bounds $\Pr(E_1)$ from below.

Proposition 4. Condition on (o_d, R). For every pixel u, $\mathbb{E}\,|f(o_d,R)(u) - o_{\text{train}}(u)| \geq e^\star(u) = \mathbb{E}\big[\inf_y \mathbb{E}[\,|y - o_{\text{train}}(u)|\,\big|\,o_d, R]\big]$, the conditional Bayes L_1 risk, attained by the conditional median; this needs no independence assumption. Summing over $u \notin V$ and bounding the terms on V below by 0 gives $\frac{1}{|\Omega|}\mathbb{E}\sum_u |\hat{o}(u) - o_{\text{train}}(u)| \geq \frac{|\Omega\setminus V|}{|\Omega|}\,\bar{e}^\star = (1-c)\,\bar{e}^\star$. For the corollary, a pixel with expected error below $e_{\min}$ cannot be a hole, since on holes the expected error is at least $e^\star(u) \geq e_{\min}$; so such pixels lie in V, a fraction at most c. The bound concerns what the deployed frame can contribute: with a stored reference R of a static scene, $e^\star$ on holes is small, and the wrapper, which fills from R, has total error $(1-c)\,e_{\text{fill}} + c\,e_V$ exactly, with e_{fill}, e_V its mean errors on holes and on V. Error on V from depth error. Let a deployed pixel back-project to X at depth z, and let the estimated depth be $z + \Delta z$, so the point used is sX, $s = 1 + \Delta z/z$. With $Y = \mathsf{R}X + t$ the point in the training camera, $\pi(Y) = Y_{xy}/Y_z$ and focal length f in pixels, $Y' = sY + (1-s)t$ and, exactly,

$$\|\Delta u\| = f\,\big\|\pi(Y') - \pi(Y)\big\| = f\,\frac{|\Delta z|}{z}\,\frac{\|t_{xy} - t_z\,\pi(Y)\|}{Y'_z}, \tag{24}$$

by direct subtraction of the two projections. The error vanishes for a pure rotation ($t = 0$) and grows with the baseline component orthogonal to the pixel's ray and with the relative depth error.

Coverage: the identity that operates, and the bound†. Under the model of Prop. 4, with $\hat{o} = f(o_d, R)$ and the error of a pixel $|\hat{o}(u) - o_{\text{train}}(u)|$ averaged over the channels:

Proposition 12 (error decomposition by coverage). For every realisation and every V with $0 < c < 1$,

$$E \;=\; \frac{1}{|\Omega|}\sum_{u\in\Omega} |\hat{o}(u) - o_{\text{train}}(u)| \;=\; (1-c)\,e_{\text{hole}} + c\,e_V, \tag{25}$$

with e_{hole} and e_V the mean errors over $\Omega \setminus V$ and over V (for $c \in \{0,1\}$ the empty term is dropped). Taking expectations, $\mathbb{E}E = (1-c)\,\mathbb{E}e_{\text{hole}} + c\,\mathbb{E}e_V$ for a fixed V.

Proof. Split the sum over Ω into the sums over V and $\Omega \setminus V$ and write each as its size times its mean; $|V|/|\Omega| = c$ and $|\Omega \setminus V|/|\Omega| = 1 - c$. Linearity of expectation gives the second statement. □

Remark (the bound of Prop. 4). Since $\mathbb{E}e_V \geq 0$ and $\mathbb{E}e_{\text{hole}} \geq \bar{e}^\star$, (25) implies $\mathbb{E}E \geq (1-c)\,\bar{e}^\star$. The bound says something only when the holes cannot be predicted: if the Bayes risk on holes is at least $e_{\min} > 0$, which is the case without side information ($R = \emptyset$, the inpainted curve of Fig. 5), the error grows at least linearly in the lost view $1 - c$. With the stored training-view reference R of a static scene $\bar{e}^\star$ is small and the bound is slack, and (25) is the operative statement. Check (results/local/fidelity.jsonl, the 264 task-placements, V the splat set at the true pose, c = filled). Recomputing $(1-c)\,e_{\text{hole}} + c\,e_V$ from the logged per-set errors reproduces the logged total with maximum absolute deviation 3.6×10^{-6} grey levels for the training-view fill and 7.6×10^{-6} for inpainting (float rounding), over c from 0 to 1.

Which set V Prop. 4 is about†. The proof of Prop. 4 never uses how V is defined: for every pixel u the expected error is at least the conditional Bayes risk $e^\star(u)$, so the bound $(1-c)\,\bar{e}^\star$ holds for any $V \subseteq \Omega$, and the choice of V only decides over which pixels $\bar{e}^\star$ is averaged. We therefore take $V := F$, the splat set of Eq. 9 evaluated at the true pose, so that the c of the proposition is exactly the implemented coverage at the true pose and $\hat{c}$ is the same quantity at the estimated pose.

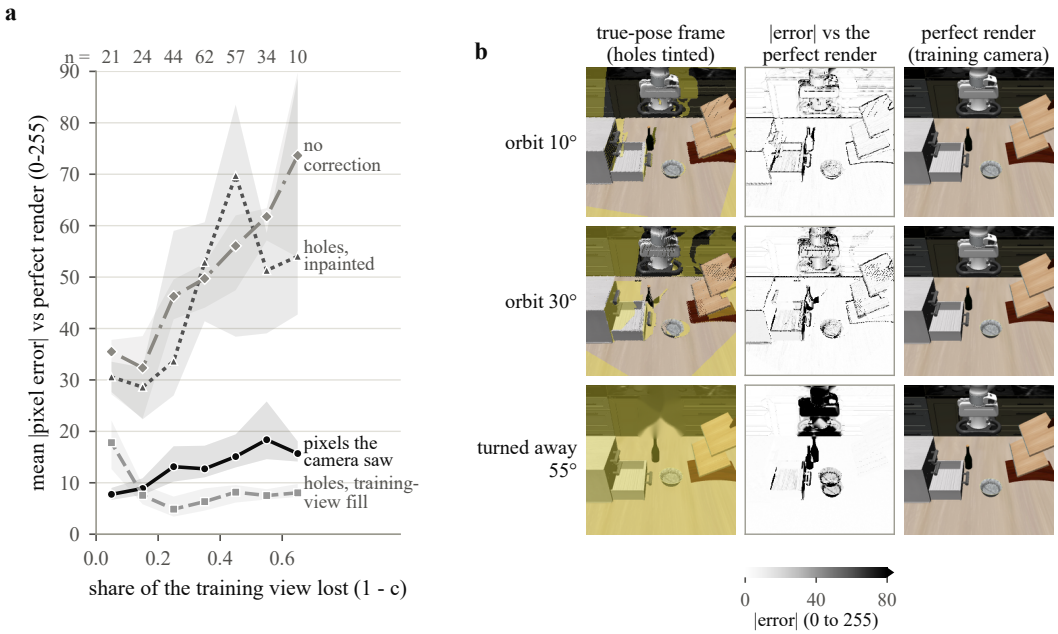


Figure 5: Where the corrected frame is wrong (Props. 4, 12). (a) Error against a perfect training-view render over the 264 task-placements (median and IQR per bin of lost view; n placements per bin above the axes; the 12 placements with more than 70% of the view lost are omitted), split into pixels the camera saw and hole pixels. With a training-view fill the holes are cheap (median 7.6, mean absolute pixel error on 0 to 255); filling them by inpainting instead costs several times more and grows with lost view; the error that grows with lost view under the training-view fill sits on the seen pixels. (b) Three cameras (rows) at the true pose, as in (a): the re-rendered frame with holes in yellow, its absolute error against the perfect render (0 to 80 shown), and the perfect render. Row 3 (turned away 55 degrees) keeps nothing of the training view at the true pose (a regime (a) omits); its measured pose is 108.9 cm off, so its self-measured coverage 0.71 is not meaningful.

Lemma 1 (splat footprint). Let S be the deployed pixels whose back-projected point (every deployed pixel is back-projected with its depth) lies more than 10^{-3} m in front of the training camera and projects within one pixel of Ω. The implementation rounds the projection to the nearest pixel centre (i, j) and writes $\{i, i+1\} \times \{j, j+1\}$. Hence $|F| \le 4|S|$, and F contains every pixel of Ω that is the nearest pixel centre to the projection of some point of S.

Proof. Each point of S writes to at most four training pixels and F is the union of these writes, so $|F| \le 4|S|$; the footprint contains the pixel whose centre is nearest the projection, and the z-buffer changes which point a pixel keeps, not whether it is filled. □

F differs from the continuous pre-image set V_{pre} (training pixels whose ray first hits a surface point visible in o_d) in three ways: a splat spills up to one pixel past depth edges ($F \not\subseteq V_{\text{pre}}$); a point visible in o_d fills a training pixel whose first surface is not visible in o_d, since the splat has no occlusion test against surfaces it never saw ($F \not\subseteq V_{\text{pre}}$); and where the deployed-to-training pixel map magnifies by more than 2 the footprints can leave gaps ($V_{\text{pre}} \not\subseteq F$). The renders (figs/renders/fid_*.npz) store only splat sets (filled at the estimated pose, filled_true at the true pose), so $|c_{\text{splat}} - c_{\text{pre}}|$ is unmeasured. What they do give is $|\hat{c} - c|$ between the two splat sets: 0.0024 and 0.0005 on the two renders with a sub-centimetre calibration (fid_high, fid_mid).

Proposition 5. With $\eta = \eta_0$ and an exact body model, the rendered and observed silhouettes are the same function of q, so $r(q)$ is a function of sensing and encoder error alone

and every realisation has $r \le \nu_{\max}$ by definition of the supremum; the contrapositive is the claim. The test is one-sided: a zero residual at one pose is consistent with a small set of poses that project the arm identically at that q, which is why the method measures with several poses once the test fires, and Proposition 13 and Lemma 2 bound that set.

The static residual as implemented†. The implemented residual (static_residual, which calls the calibration score with one record) renders the model mask at the configuration the encoders report at the moment of the check, $\tilde{q}$, not at a stored rest pose: $r(\tilde{q};\eta_0)$ of Eq. 12 with M_{obs} the robot mask in the deployed frame and $M(\tilde{q},\eta_0)$ the model mask at the training camera. Joint offsets larger than 0.05 rad are first servoed back to the training start, so the check sees the arm near q_{nom} in any case. Prop. 5 is therefore about encoder readings.

Corollary 1 (encoder error enters the floor). Treat the masks as continuous regions (rasterisation can add up to one pixel). Let the encoder reading be $\tilde{q} = q + \Delta q$, with sensing otherwise exact and $\eta = \eta_0$, and let the model boundary move with q so that every boundary point moves by at most $d = \|J_q \Delta q\|_{2,\infty} + o(\|\Delta q\|)$ pixels, J_q the Jacobian of the boundary points (pixels) in the joint configuration at the rest pose q and $\|\cdot\|_{2,\infty}$ the largest per-point displacement. With P the boundary length and A the area of the mask in pixels,

$$r(\tilde{q};\eta_0) \le \kappa_q \, \|J_q \Delta q\|_{2,\infty} + o(\|\Delta q\|), \qquad \kappa_q = 1 + \frac{4P}{A+1}, \tag{26}$$

so bounded encoder error contributes at most $\kappa_q \sup \|J_q \Delta q\|_{2,\infty}$ to $\nu_{\max}$, to first order.

Proof. With $\Delta q = 0$ the two masks coincide. After the displacement the Hausdorff distance between the two boundaries is at most d, so every nearest-boundary distance in either mean of Eq. 12 is at most d and the chamfer term is at most d. The area changes by at most the swept band, $|\Delta A| \le Pd + O(d^2)$, and $|\log((A+\Delta A+1)/(A+1))| = |\Delta A|/(A+1) + O(\Delta A^2)$, so the size term is at most $4Pd/(A+1) + O(d^2)$. Sum the two. □

In the simulator the encoder reading equals the true configuration, so the corollary is not exercised by these rows. Check (the 170 $\pi_{0.5}$ graded-sweep units of the motion-free arm, one row each). 145 read exactly 0 and 25 read above 0, at most 0.020 (offset 0.20 on task 1442; 0.019, offset 0.35 on task 1167). All 25 nonzero units had the joint servo run immediately before the check (state_applied): 11 are joint-offset units and 14 are the non-camera units of base task 1101, all at 0.0066, whose untouched start is 0.071 rad from the training start (above the 0.05 rad servo threshold); 19 servoed units read exactly 0. The rows therefore show that nonzero residuals occur only after the servo. A plausible source, not verified, is that the servo hands the arm over still moving, so the frame and the re-render at the reported $\tilde{q}$ differ by one control step.

Proposition 13 (detection radius). Let J_q be the Jacobian of the one-pose self-observation in η at the rest pose q, and suppose that near η_0 the residual satisfies $r(q;\eta) \ge \kappa \|J_q(\eta - \eta_0)\| - \nu$ for a constant $\kappa > 0$ converting self-observation units to pixels, with ν the sensing floor (the largest unshifted residual, as in Prop. 18). Then every shift with $\|J_q(\eta - \eta_0)\| > (\tau + \nu)/\kappa$, in particular every shift with $\|\eta - \eta_0\| > (\tau + \nu)/(\kappa\, \sigma_{\min}(J_q))$, is detected by the rule $r > \tau$, and every undetected shift lies in the ellipsoid $\kappa \|J_q \delta\| \le \tau + \nu$.

Proof. $r \ge \kappa \|J_q \delta\| - \nu > \tau$ when $\kappa \|J_q \delta\| > \tau + \nu$, and $\|J_q \delta\| \ge \sigma_{\min}(J_q) \|\delta\|$ gives the sufficient condition on $\|\delta\|$. Conversely $r \le \tau$ implies $\kappa \|J_q \delta\| \le \tau + \nu$. □

The rows do not identify κ; the lemma below replaces it by one logged residual per unit and a shape assumption.

Detection radius from the rows†.

Lemma 2 (radius from one logged shift). Fix a unit u and a one-parameter family of shifts (orbit azimuth $\theta \ge 0$, degrees), with geometric residual $r_u(\theta)$ and observed residual $r_u(\theta) + e$,

$|e| \le \nu$. If $\theta \mapsto r_u(\theta)/\theta$ is nonincreasing on $(0, \theta_1]$, then $r_u(\theta) \ge \theta\, r_u(\theta_1)/\theta_1$ on $(0, \theta_1]$, and the rule $r > \tau$ fires for every

$$\theta \in \Big(\frac{(\tau + \nu)\,\theta_1}{r_u(\theta_1)},\ \theta_1 \Big], \tag{27}$$

an interval that is nonempty iff $r_u(\theta_1) > \tau + \nu$. A sufficient condition for the premise is that r_u is concave on $[0, \theta_1]$ with $r_u(0) = 0$.

Proof. For $\theta \le \theta_1$ the premise gives $r_u(\theta)/\theta \ge r_u(\theta_1)/\theta_1$. The rule fires if $r_u(\theta) - \nu > \tau$, which holds when $\theta\, r_u(\theta_1)/\theta_1 > \tau + \nu$, i.e. on (27). For the sufficient condition, write $\theta = \lambda\theta'$ with $0 < \theta < \theta' \le \theta_1$ and $\lambda \in (0, 1)$: concavity gives $r_u(\theta) \ge \lambda r_u(\theta') + (1-\lambda) r_u(0) = \lambda r_u(\theta')$, i.e. $r_u(\theta)/\theta \ge r_u(\theta')/\theta'$. □

Check (results/local/static_residual.jsonl, simulator masks, one rest pose). The 10 sweep tasks have orbit rows at $\theta = 10, 20, 30, 45°$ and read exactly 0 unshifted, so $r_u(0) = 0$ and $\nu = 0$. With $\theta_1 = 10°$ and $\tau = 1.5$, (27) gives a radius of $1.5 \times 10/r_u(10)$: 4.83° to 7.77° over the tasks (median 5.89°), the worst task (1297) at $1.5 \times 10/1.929 = 7.8°$. The logged true shifts at 10° (13.6 to 15.2 cm) put the orbit's horizontal radius at 78 to 87 cm, so the radius is a camera-centre displacement of 7.1 to 11.0 cm (worst task 10.6 cm) together with the same rotation; one degree of orbit moves the centre 1.4 to 1.5 cm, so the measured tolerance of about 1 cm and 1° is below one degree of orbit. With the sweep floor $\nu = 0.020$ the worst radius is 7.9°; with the learned-mask floor $\nu = 0.906$ (plug-in, since r_u under learned masks is not logged) the interval is empty on the worst task ($2.406 \times 10/1.929 = 12.5° > \theta_1$) and nonempty on 6 of 10. The premise. r_u is nondecreasing on the grid on all 10 tasks, but $r_u(\theta)/\theta$ is nonincreasing over $10, 20, 30, 45°$ on only 5 of 10; on 4 (1208, 1297, 1412, 1442, including the worst) it rises from 10° to 20°, and on 1101 from 30° to 45°. The premise concerns $(0, 10°]$, where there are no rows; a ratio that rises just above 10° suggests a residual that starts flatter than linear, in which case the premise can fail below 10° and the true radius can be larger than (27). The radius is therefore a conditional statement, and between the tolerance (under 1° of orbit) and about 8° the check is not shown to fire. A policy-free Colab run at 1, 2, 4, 6, 8 and 10° of orbit, which tests the premise directly, is not yet run.

Proposition 6. Write $w_k = p_k - g^\star$, $r_k = \|w_k\|$, $a_k = a(p_k) = Aw_k$, $u_k = a_k/\|a_k\|$, $P_k = I - u_k u_k^\top$ and $M = \sum_k P_k$; M is positive definite unless all a_k are parallel, and the least-squares point solves $M\hat{x} = \sum_k P_k p_k$ with residual $\sum_k \|P_k(\hat{x} - p_k)\|^2$. The parameter of the point of ray $p_k + tu_k$ closest to $\hat{x}$ is $t_k = u_k^\top(\hat{x} - p_k)$. (i) If $A = \mu I$, $a_k = \mu w_k$ and the ray through p_k passes through $g^\star$ at $t_k = -\mathrm{sign}(\mu)\, r_k$, so all rays meet at $g^\star = \hat{x}$, the residual is zero whatever the camera, and $g^\star$ is behind probe k ($t_k < 0$) iff $\mu > 0$. (ii) If $A = \mu I + E$, $\|E\| \le \varepsilon|\mu|$, $\varepsilon < 1$: $P_k a_k = 0$ gives $\mu P_k w_k = -P_k E w_k$, so $\|P_k(g^\star - p_k)\| \le \varepsilon r_k$. The normal equations give $M(\hat{x} - g^\star) = \sum_k P_k w_k$, so $\|\hat{x} - g^\star\| \le \varepsilon \sum_k r_k/\lambda_{\min}(M) \le K\varepsilon r_{\max}/\lambda_{\min}(M)$. For $\mu > 0$, $u_k^\top(g^\star - p_k) = -w_k^\top A^\top w_k/\|Aw_k\| \le -\frac{(1-\varepsilon)}{(1+\varepsilon)} r_k$, hence $t_k \le \|\hat{x} - g^\star\| - \frac{1-\varepsilon}{1+\varepsilon} r_k < 0$ whenever $K\varepsilon r_{\max}/\lambda_{\min}(M) < \frac{1-\varepsilon}{1+\varepsilon} r_k$ (symmetrically for $\mu < 0$). The residual is at most $\sum_k (\varepsilon r_k + \|\hat{x} - g^\star\|)^2$, which vanishes with E for any camera: it measures the anisotropy of the field, not miscalibration. (iii) $\mathrm{tr}\, A$ is the sum of the eigenvalues, so $\mathrm{tr}\, A > 0$ forces an eigenvalue with positive real part, and the equilibrium $g^\star$ of $\dot{p} = A(p - g^\star)$ is not asymptotically stable; equivalently the flow's divergence is $\mathrm{tr}\, A > 0$ and it expands volumes (Liouville). A positive definite symmetric part implies $x^\top A x > 0$ for $x \ne 0$, so A is invertible and $g^\star$ is defined; it also implies $\mathrm{tr}\, A > 0$. Check. $\hat{A}$ is fit by least squares, $a(p_k) = a_0 + A(p_k - p_1)$ over the 7-probe cross, on the raw frame, from results/drive_vihan/sweep/smc_*.jsonl (72, 72, 96 units for OFT, $\pi_{0.5}$, OpenVLA): $\mathrm{tr}\, \hat{A} > 0$ on 97%, 97%, 96%; $\lambda_{\min}(\mathrm{sym}\, \hat{A}) > 0$ on 15%, 0%, 1%; the triangulated point behind the centre probe ($t_1 < 0$) on 94%, 78%, 54%. The rows support (iii); the isotropic picture of (i) and (ii) holds only roughly, on OFT.

Invariances of the probe field and a discrete-time form of Prop. 6†. The coordinates of p_k and a_k are stated in Appendix B.1 (fingerprint distance).

Proposition 14 (Prop. 6, parts (i) and (iii) restated). (i) Given the probe positions, the triangulation residual is a function of the fitted field (a_0, A) only (equivalently of A and $g^\star$), unchanged by a positive rescaling of the field, and zero whenever $A = \mu I$ whatever a_0; the camera can register only through (a_0, A). (iii) If $\operatorname{tr} A > 0$, $g^\star$ is not an attractor of $\dot{p} = a(p)$, and for the discrete update $p_{t+1} = p_t + \alpha\, a(p_t)$ it is not asymptotically stable for any step $\alpha > 0$. (iv) The sign of $\operatorname{tr} A$ is invariant under similarity $A \mapsto SAS^{-1}$ (a common change of coordinates of positions and actions) but not under a change of action coordinates alone, $a \mapsto Da + b$ with D positive diagonal (e.g. q_{01}/q_{99} normalisation), which maps $A \mapsto DA$; for $\operatorname{tr} A > 0$ it is invariant under every such D iff $A_{ii} \geq 0$ for all i.

Proof. (i) By the proof of Prop. 6 the rays are determined by the probe positions, read from the encoders and so independent of the camera, and the directions $u_k = a(p_k)/\|a(p_k)\| = Aw_k/\|Aw_k\|$, $w_k = p_k - g^\star$, which depend on the field only through (a_0, A) and are unchanged when a is multiplied by a positive scalar; for $A = \mu I$ the residual is zero by Prop. 6(i). (iii) $\operatorname{tr} A$ is the sum of the eigenvalues, so some eigenvalue λ has $\operatorname{Re}\lambda > 0$. The error $e_t = p_t - g^\star$ obeys $e_{t+1} = (I + \alpha A)e_t$, whose eigenvalue $1 + \alpha\lambda$ has $|1 + \alpha\lambda|^2 = (1 + \alpha \operatorname{Re}\lambda)^2 + \alpha^2(\operatorname{Im}\lambda)^2 > 1$ for every $\alpha > 0$; an initial error with a nonzero component along the corresponding (generalised) eigenvector grows, so $g^\star$ is not asymptotically stable. (iv) $\operatorname{tr}(SAS^{-1}) = \operatorname{tr} A$. With $a' = Da + b$, $a'(p) = (Da_0 + b) + DA(p - p_1)$, so $A' = DA$ and $\operatorname{tr}(DA) = \sum_i d_i A_{ii}$, which is positive for every $d > 0$ iff all $A_{ii} \geq 0$ and one is positive (take d concentrated on a negative entry otherwise); with $\operatorname{tr} A > 0$ some entry is positive. □

Check (results/drive_vihan/sweep/smc_*.jsonl, raw frame, $\hat{A}$ by least squares of $a(p_k) = a_0 + A(p_k - p_1)$ on the 7 probes, 12 coefficients from 21 numbers, 9 residual degrees of freedom; $\mathrm{SE}(\operatorname{tr}\hat{A})^2 = \hat{s}^2 \sum_{i=1}^{3} [(X^\top X)^{-1}]_{ii}$ over the three slope coefficients, $\hat{s}^2 = \mathrm{RSS}/9$; this treats the residuals of the three action axes as having one common variance and no correlation across axes). $\operatorname{tr}\hat{A} > 0$ on 70 of 72, 70 of 72 and 92 of 96 units (OFT, $\pi_{0.5}$, OpenVLA), reproducing the 97%, 97%, 96% of Sec. 4.5; but $\operatorname{tr}\hat{A} > 2\,\mathrm{SE}$ on only 2 of 72 (3%), 26 of 72 (36%) and 25 of 96 (26%), and no unit has $\operatorname{tr}\hat{A} < -2\,\mathrm{SE}$ (median t-ratio 1.23, 1.59, 1.45). The positive divergence is therefore a population statement: individually most fits do not resolve it. The sign is robust to any per-axis rescaling of the actions (all $\hat{A}_{ii} > 0$) on 39 of 72, 23 of 72 and 32 of 96 units; the q_{01}/q_{99} statistics are not stored locally, so the normalised-coordinate share is not reported. Metric units differ from command units by a positive scalar and give the same shares.

Divergence is a statement about the probed region†. Assumptions. (S) Stationarity: the probe actions are samples of one time-invariant field $a(p)$ of the end-effector position, the same whatever the time, history or chunk index at which the policy is queried. (R) Realised motion: the commanded action is the realised velocity up to a positive scalar (the operational-space controller tracks the command), so the field of commands is the field of motions. (L) The field is affine on the probed ball $B = \{p : \|p - p_1\| \leq R_{\mathrm{probe}}\}$, $R_{\mathrm{probe}} = 0.10$ m, which contains the 7 probes (±10 cm in x, y, +8 and −6 cm in z).

Proposition 15 (no equilibrium in the probed ball). Under (S), (R), (L), with $a(p) = a_0 + A(p - p_1)$ on B: if A is invertible and $\|A^{-1}a_0\| > R_{\mathrm{probe}}$, or A is singular and a has no zero in B, then (i) the linear model has no equilibrium in B, and every forward and every backward trajectory of $\dot{p} = a(p)$ that starts in B leaves B in finite time; (ii) for a measurable $S \subseteq B$ and the flow φ_t, $\operatorname{vol}(\varphi_t(S)) = e^{t \operatorname{tr} A} \operatorname{vol}(S)$ while $\varphi_t(S) \subseteq B$, and one step of $p \mapsto p + \alpha a(p)$ multiplies volume by $|\det(I + \alpha A)| = 1 + \alpha \operatorname{tr} A + O(\alpha^2)$. So on B, $\operatorname{tr} A > 0$ says the fitted local field expands volume during its transit of the probed region; it says nothing about an attractor, which the model places outside B.

Proof. (i) The unique zero of a is $g^\star = p_1 - A^{-1}a_0$, at distance $\|A^{-1}a_0\| > R_{\mathrm{probe}}$ from p_1. Suppose a forward trajectory stays in B for all $t \geq 0$. Since a is affine and B convex, the time average $\bar{p}_T = T^{-1} \int_0^T p(t)\, dt$ lies in B and $a(\bar{p}_T) = T^{-1} \int_0^T a(p(t))\, dt = (p(T) - p(0))/T$, whose norm is at most $2R_{\mathrm{probe}}/T \to 0$. B is compact, so a subsequence $\bar{p}_{T_k} \to \bar{p} \in B$ with $a(\bar{p}) = 0$ by continuity, a zero in B: contradiction. The backward case is the same

argument for $\dot{p} = -a(p)$. (ii) The divergence of an affine field is $\mathrm{tr}A$, and Liouville's formula gives $\frac{d}{dt}\mathrm{vol}(\varphi_t(S)) = \int_{\varphi_t(S)} \mathrm{tr}A = \mathrm{tr}A\ \mathrm{vol}(\varphi_t(S))$ while the image stays where the model holds. The discrete map is affine with linear part $I + \alpha A$, whose determinant is $\prod_i(1 + \alpha\lambda_i) = 1 + \alpha\,\mathrm{tr}A + O(\alpha^2)$. □

Check (results/drive_vihan/sweep/smc_*.jsonl, raw frame, $(\hat{a}_0, \hat{A})$ by least squares on the 7 probes as in Prop. 14). The distance $\|\hat{A}^{-1}\hat{a}_0\|$ from the centre probe to the fitted equilibrium has median 1.60 m (quartiles 0.58, 2.41) on OFT, 0.31 m (0.25, 0.64) on $\pi_{0.5}$ and 0.21 m (0.11, 2.17) on OpenVLA; it is within $R_{\mathrm{probe}} = 0.10$ m on 0 of 72, 0 of 72 and 20 of 96 units. On 220 of 240 units the premise holds, and the fitted equilibria (the $g^\star$ of Prop. 6) are extrapolations, mostly 0.2 m to several metres from the centre probe; the probes do not reach the equilibria that stable dynamical-system policies (Beik-Mohammadi et al., 2024) would contract to, so the rows neither support nor contradict such contraction.

A test for the sign of the divergence†.

Lemma 3 (sign test under symmetric error). Let $\mathrm{tr}\hat{A}_u = \mathrm{tr}A_u + \varepsilon_u$ for units $u = 1, \ldots, n$. (a) If the ε_u are independent and each is symmetric about 0, and $\mathrm{tr}A_u \le 0$ for every u, then $\Pr(\mathrm{tr}\hat{A}_u > 0) \le \frac{1}{2}$ for each u and $N_+ = \#\{u : \mathrm{tr}\hat{A}_u > 0\}$ is stochastically smaller than $\mathrm{Bin}(n, \frac{1}{2})$, so $p = \Pr(\mathrm{Bin}(n, \frac{1}{2}) \ge N_+)$ is a valid one-sided p value. (b) If exactly m units have $\mathrm{tr}A_u > 0$, N_+ is stochastically smaller than $m + \mathrm{Bin}(n - m, \frac{1}{2})$, and $m_L = \min\{m : \Pr(\mathrm{Bin}(n - m, \frac{1}{2}) \ge N_+ - m) > \alpha\}$ is a one-sided $1 - \alpha$ lower confidence bound on m. (c) If the units fall into G independent clusters with arbitrary dependence inside a cluster but a cluster error vector whose law is invariant under $\varepsilon \mapsto -\varepsilon$, and $\mathrm{tr}A_u \le 0$ for every u, then each cluster's majority sign is positive with probability at most $\frac{1}{2}$, and the number of clusters with a positive majority is stochastically smaller than $\mathrm{Bin}(G, \frac{1}{2})$.

Proof. (a) $\mathrm{tr}\hat{A}_u > 0$ requires $\varepsilon_u > -\mathrm{tr}A_u \ge 0$, so $\Pr(\mathrm{tr}\hat{A}_u > 0) \le \Pr(\varepsilon_u > 0) \le \frac{1}{2}$ by symmetry. Independent Bernoulli variables with success probabilities $\pi_u \le \frac{1}{2}$ are coupled below independent Bernoulli($\frac{1}{2}$) variables ($\mathbf{1}[U_u \le \pi_u] \le \mathbf{1}[U_u \le \frac{1}{2}]$ for uniform U_u), so their sum is stochastically below $\mathrm{Bin}(n, \frac{1}{2})$. (b) The same coupling with $\pi_u \le 1$ on the m expanding units and $\pi_u \le \frac{1}{2}$ on the others; the set of m not rejected at level α contains the true m with probability at least $1 - \alpha$, and m_L is its smallest element (the tail probability is nondecreasing in m). (c) Within a cluster, $\mathrm{tr}\hat{A}_u > 0$ implies $\varepsilon_u > 0$, so a positive majority of $\mathrm{tr}\hat{A}$ implies a positive majority of ε; by the sign invariance this has the same probability as a negative majority of ε, and the two events are disjoint, so each has probability at most $\frac{1}{2}$. Independence across clusters and the coupling of (a) finish. □

Symmetry holds for least-squares errors under Gaussian noise and a linear field. The standard error of Prop. 14 uses $\hat{s}^2 = \mathrm{RSS}/9$, which includes any lack of fit: if the field is not linear on the probes the SE is inflated (conservative), but $\mathrm{tr}\hat{A}$ then estimates the trace of the best affine approximation over the probe design rather than a derivative, so the SE is a sampling SE only if the field is linear. Check (the rows of Prop. 14). Unit level (a): 70 of 72, 70 of 72 and 92 of 96 positive give one-sided $p = 5.6 \times 10^{-19}$, 5.6×10^{-19} and 4.4×10^{-23} (OFT, $\pi_{0.5}$, OpenVLA); by (b) at least 62 of 72, 62 of 72 and 81 of 96 units (0.86, 0.86, 0.84) have $\mathrm{tr}A > 0$ at 95%. But the units are 24 conditions on each of only 3, 3 and 4 base tasks and share them, so (a) and (b) assume more independence than the design gives. Task level (c): every base task has a positive majority (3 of 3, 3 of 3, 4 of 4), $p = 0.125$, 0.125 and 0.0625: with 3 or 4 clusters no task-level test can reach 0.05, and the positive divergence is a unit-level population statement on few tasks.

B.3 The coverage estimate and the product form

Lemma 4 (coverage from an estimated pose). If the reprojection's filled set moves with the pose so that its boundary ∂V, of length $|\partial V|$ pixels, is displaced by at most $L\|\hat{\eta} - \eta\|$ pixels (locally L-Lipschitz), then $|\hat{c} - c| \le L\,|\partial V|\,\|\hat{\eta} - \eta\|/|\Omega| + O(\|\hat{\eta} - \eta\|^2)$.

Proof. $\hat{c} - c = (|\hat{V}| - |V|)/|\Omega|$, and $|\hat{V} \triangle V|$ is at most the area swept by ∂V moving at most $L\|\hat{\eta} - \eta\|$ along its normal, which is $L|\partial V|\,\|\hat{\eta} - \eta\|$ to first order; the second order term is the curvature correction of the swept band. □

On the 264 policy-free task-placements (88 camera placements × 3 base tasks), where both are logged (results/local/envelope.jsonl: $\hat{c}$ from the self-calibrated pose; results/local/fidelity.jsonl: c from the true pose), the median $|\hat{c} - c|$ is 0.004 (90th percentile 0.12, maximum 0.71 at gross calibration failures).

Why a product†. Repair is helped by two events: E_1, the calibration error falls below the policy's tolerance, whose probability Prop. 11 bounds below by an increasing function of ident (the probability itself need not be monotone); and E_2, the restored fraction exceeds what the policy needs, which Prop. 12 makes harder as c falls (the coverage-repair relation itself is empirical, Fig. 2b). Without an independence assumption the joint probability is only constrained by the Fréchet bounds, and repair can also happen outside $E_1 \cap E_2$:

$$\begin{gathered}\max(0, P_1 + P_2 - 1) \le \Pr(E_1 \cap E_2) \le \min(P_1, P_2),\\ \Pr(\text{repair}) = \Pr(E_1 \cap E_2)\,\Pr(\text{success} \mid E_1, E_2) + \Pr(\text{repair},\ \overline{E_1 \cap E_2}),\end{gathered} \tag{28}$$

with $P_i = \Pr(E_i)$ (the bounds follow from $\Pr(E_1 \cup E_2) \le 1$ and $E_1 \cap E_2 \subseteq E_i$; the decomposition is the law of total probability). The product ident × invert is therefore a heuristic ranking, not a probability; it equals $P_1 P_2$ only under independence and only if its factors are those probabilities. Check (results/local/coverage_units.jsonl, the 98 raw failures of Fig. 2b; $E_1 = \{\text{error} \le 2\text{ cm}\}$, $E_2 = \{\hat{c} \ge \text{median } 0.631\}$). $E_1 \cap E_2$: 27, E_1 only 22, E_2 only 22, neither 27; $P_1 = P_2 = 0.50$, $\Pr(E_1 \cap E_2) = 0.276$ against the independence value 0.25 (Fréchet range $[0, 0.50]$); Fisher exact $p = 0.42$ (odds ratio 1.51). On all 234 camera tasks with both logged, $p = 0.43$ (odds ratio 1.27). So the rows show no detectable association between the two events, and do not establish independence either. The decomposition (28) on the 98 failures: $\Pr(\text{repair}) = 0.551 = 0.276 \times 0.926 + 0.296$; repair given both events is 0.93, but more than half of the repairs fall outside $E_1 \cap E_2$ at these thresholds, so the two events are not necessary for repair. The Fréchet upper bound as a score (Table 1 units): min(ident, invert) has AUROC 0.828 [0.72, 0.92] (OFT, 65 units), 0.795 [0.68, 0.90] ($\pi_{0.5}$, 72) and 0.678 [0.53, 0.82] (OpenVLA, 60), against 0.823, 0.809, 0.675 for the product; paired differences, product minus minimum, −0.005 [−0.03, 0.02], +0.014 [−0.01, 0.04], −0.003 [−0.02, 0.01] (score_baselines.py--ext, unit bootstrap). The combination rule does not matter on these units; the ranking is carried by invert. The product beats ident alone on OFT and $\pi_{0.5}$ (intervals exclude 0) and numerically on OpenVLA (+0.16 [−0.00, 0.32]), and ties invert alone (Table 1; paired difference on $\pi_{0.5}$ −0.01 [−0.10, 0.07]).

Where ident moves the AUROC†.

Lemma 5 (pair decomposition). Let $v_i = \text{invert}_i \ge 0$, $s_i = \text{ident}_i v_i$ ($\text{ident}_i \in [0, 1]$), $U_1 = \{i : \text{ident}_i = 1\}$, $U_2 = \{i : \text{ident}_i < 1\}$ and $h(a, b) = \mathbf{1}[a > b] + \frac{1}{2}\mathbf{1}[a = b]$. On the same P positives and N negatives,

$$\widehat{A}(\text{recover}) - \widehat{A}(\text{invert}) = \frac{1}{PN} \sum_{\substack{i:\, y_i = 1,\ j:\, y_j = 0 \\ \{i,j\} \cap U_2 \ne \emptyset}} \big[h(s_i, s_j) - h(v_i, v_j)\big], \tag{29}$$

and the terms with $i \in U_2$, $j \in U_1$ are ≤ 0, those with $i \in U_1$, $j \in U_2$ are ≥ 0; pairs inside U_2 can take either sign.

Proof. Subtract the two instances of (4). On U_1, $s_i = v_i$, so a pair with both units in U_1 contributes 0. For $i \in U_2$ (a positive) and $j \in U_1$, $s_i \le v_i$ and $s_j = v_j$, and h is nondecreasing in its first argument, so the term is ≤ 0; for $i \in U_1$ and $j \in U_2$ (a negative), $s_j \le v_j$ and h is nonincreasing in its second argument, so it is ≥ 0. □

So ident helps only by demoting negatives and hurts only by demoting positives. Check (score_baselines.py –pairs, sweep units with both factors defined; the differences reproduce Table 1 and App. D). OFT: 65 units, $|U_2| = 32$ (11 look-away, 15 of the 32 repaired);

$\widehat{A}(\text{recover}) - \widehat{A}(\text{invert}) = 0.823 - 0.831 = -0.008$, of which demoted negatives $+0.038$, demoted positives -0.029, pairs within U_2 -0.016; pairs touching a look-away unit $+0.018$, the others -0.026. $\pi_{0.5}$: 72 units, $|U_2| = 41$ (11 look-away, 17 repaired); $-0.010 = +0.045$, -0.030, -0.026; look-away -0.005, others -0.005. OpenVLA: 60 units, $|U_2| = 26$ (3 look-away, 11 repaired); $+0.004 = +0.039$, -0.041, $+0.006$; look-away $+0.006$, others -0.002. On every family ident gains about 0.04 by demoting negatives and gives back about as much by demoting repaired units with ident < 1 (15, 17 and 11 of them), which is why the product ties invert alone; on OFT its net gain is on the look-away units and its net loss elsewhere.

B.4 Further propositions

Label contamination and the AUROC†.

Proposition 16 (contaminated positives). Let the P positive units split into $P - m$ genuine positives $\mathcal{G}$ and m contaminated positives $\mathcal{C}$, and let $\widehat{A}_\mathcal{G}$, $\widehat{A}_\mathcal{C}$ be (4) computed on $\mathcal{G}$, respectively $\mathcal{C}$, against all N negatives. Then, exactly,

$$\widehat{A} = \frac{P-m}{P}\,\widehat{A}_\mathcal{G} + \frac{m}{P}\,\widehat{A}_\mathcal{C}, \qquad \widehat{A}_\mathcal{G} - \widehat{A} = \frac{m}{P}\,\big(\widehat{A}_\mathcal{G} - \widehat{A}_\mathcal{C}\big). \tag{30}$$

(i) If every contaminated unit has the smallest attainable score (invert $= 0$, so recover $= 0$) and z_0 of the negatives also score 0, then $\widehat{A}_\mathcal{C} = z_0/(2N)$ and $\widehat{A} = \widehat{A}_\mathcal{G} - \frac{m}{P}\big(\widehat{A}_\mathcal{G} - \frac{z_0}{2N}\big)$. (ii) If instead the contaminating mechanism acts independently of the score, i.e. the contaminated units are would-be negatives relabelled at random, so their scores are independent draws from the negatives' score distribution F_0 and independent of the observed negatives, then $\mathbb{E}\,\widehat{A}_\mathcal{C} = 1/2$ and $\mathbb{E}\,\widehat{A} = \mathbb{E}\,\widehat{A}_\mathcal{G} - \rho\,(\mathbb{E}\,\widehat{A}_\mathcal{G} - \frac{1}{2})$ with $\rho = m/P$ (expectations over all scores; conditional on the observed negatives $\widehat{A}_\mathcal{C}$ need not have mean $1/2$): when $\mathbb{E}\,\widehat{A}_\mathcal{G} \geq 1/2$, contamination of this kind pulls the AUROC toward chance in expectation, never below it. An observed $\widehat{A}_\mathcal{C}$ well below $1/2$ is therefore evidence that the contamination is concentrated at low scores, not merely present.

Proof. The double sum in (4) runs over the PN positive-negative pairs; splitting the outer sum over $\mathcal{G} \cup \mathcal{C}$ gives $PN\,\widehat{A} = (P-m)N\,\widehat{A}_\mathcal{G} + mN\,\widehat{A}_\mathcal{C}$, which is (30). (i) A contaminated unit with $s = 0 \leq s_j$ for all j wins no pair and ties the z_0 negatives with $s_j = 0$, so each contributes $z_0/2$ of N pairs. (ii) If s_i for $i \in \mathcal{C}$ has the negatives' law F_0 and is independent of them, $\Pr(s_i > s_j) + \frac{1}{2}\Pr(s_i = s_j) = 1/2$ by exchangeability of two independent draws from F_0, unconditionally over both draws; take expectations in (30), which is linear. □

Why the observed contamination sits at the bottom. A repair produced by the measurement's arm motion is visible only on units where no channel fired; where a channel fired it is indistinguishable from a genuine repair and is counted as one. A channel does not fire where there is nothing it can invert: on scene content (invert $= 0$ by definition) and where it abstains, and the photometric channel abstains when the body's shading statistic has moved, which is what lowers $\text{invert}_{\text{light}}$. The units that can be identified as contaminated are therefore selected toward low invert. Case (i) describes four of the five OpenVLA units exactly; case (ii) bounds what an unselected mechanism could do.

Check (results/drive_vihan/sweep/episodes_openvla.*.jsonl merged with plus_common.load_shards, scores from scores_v2.jsonl via sweep_analysis.recoverability, 24 standard conditions; a unit is contaminated when its wrapped row has the four flags state_applied, cam_applied, photo_applied and denoise_applied all false). OpenVLA: $P = 24$, $N = 36$, $m = 5$; $\widehat{A}_\mathcal{G} = 558/684 = 0.816$. Four contaminated units are background conditions with recover $= 0$ and $z_0 = 5$ negatives also score 0, contributing $4 \times 5/2 = 10$ pair-wins; the fifth (grazing light, recover $= 0.599$) beats 15 of the 36 negatives, with no ties. So $\widehat{A}_\mathcal{C} = 25/180 = 0.139$ and (30) gives $(558 + 25)/864 = 0.675$, the reported value, to the digit; the drop $0.816 - 0.675 = 0.141 = \frac{5}{24}(0.816 - 0.139)$. $\pi_{0.5}$: $P = 37$, $N = 35$, $m = 1$ (background, recover $= 0$), $z_0 = 3$, so $\widehat{A}_\mathcal{C} = 3/70$ and $(1046.5 + 1.5)/1295 = 0.809$ from $\widehat{A}_\mathcal{G} = 0.831$. OFT: $m = 0$, 0.823 unchanged. Under (ii) the OpenVLA value would have

been $0.816 - \frac{5}{24}(0.316) = 0.750$ in expectation; the observed 0.675 is below that because $\widehat{A}_{\mathcal{C}} = 0.139 < 1/2$, as the selection argument predicts (the 0.750 plugs the observed $\widehat{A}_{\mathcal{G}}$ into the expectation, so it is a plug-in value, not a bound). The symmetric exclusion of Sec. 4.1 drops, on every family, all units with nothing applied, whatever their outcome (OpenVLA: 11 such units, 5 of them repaired), which is why it moves OFT and $\pi_{0.5}$ as well.

The fingerprint distance under a small pose error†.

Proposition 17 (local form of the fingerprint distance). Fix a task and frame and let $A(\delta) \in \mathbb{R}^{K\times 3}$ be the probe actions when the correction uses the pose $\eta+\delta$, $\delta \in \mathbb{R}^6$ (cm and degrees), with the sampler seed fixed so that $A(0) = A_0$ exactly. Assume A is twice differentiable on a ball $\|\delta\| \le r$ with $\|D^2 A\| \le L$ there (operator norm into the Frobenius norm), $A_0 \neq 0$, and write $\mathcal{J} = DA(0)$. (The implemented reprojection rounds to pixel centres, so the computed A is piecewise constant under sub-pixel pose changes; the assumption describes the response above that quantum, and parts (i) and (ii) do not use it.) Then for $\|\delta\| \le r$

$$d\big(A(\delta), A_0\big) = \frac{\|\mathcal{J}\delta\|_F}{\|A_0\|_F} + e(\delta), \qquad |e(\delta)| \le \frac{L\,\|\delta\|^2}{2\,\|A_0\|_F}. \tag{31}$$

Along a unit direction u the coefficient is $\kappa_u = \|\mathcal{J}u\|_F/\|A_0\|_F$, the relative sensitivity of the probe actions to pose error along u; $d(tu) = \kappa_u t + O(t^2)$ and, for $\kappa_u > 0$, the tolerance $t^\star_u = \inf\{t : d(tu) > \tau\}$ is $\tau/\kappa_u + O(\tau^2)$. Moreover: (i) if every reference probe action $a_{0,k} \neq 0$, with d_k the single-probe distances and $w_k = \|a_{0,k}\|^2/\|A_0\|_F^2$, $d^2 = \sum_k w_k d_k^2$, so $\min_k d_k \le d \le \max_k d_k$ and $\kappa_u^2 = \sum_k w_k \kappa_{u,k}^2$; (ii) if $t \mapsto d(tu)/t$ is nonincreasing on $(0, T]$, then $t^\star_u \le \tau s/d(su)$ for every $s \le T$ with $d(su) \ge \tau$, and, where the expansion holds, $\kappa_u \ge d(tu)/t$ for every $t \le T$.

Proof. Taylor's theorem with integral remainder gives $A(\delta) - A_0 = \mathcal{J}\delta + R(\delta)$, $\|R(\delta)\|_F \le \frac{L}{2}\|\delta\|^2$. By the triangle inequality $\big|\,\|A(\delta) - A_0\|_F - \|\mathcal{J}\delta\|_F\big| \le \|R(\delta)\|_F$; dividing by $\|A_0\|_F$ (which exceeds the guard 10^{-9}) gives (31). Setting $\delta = tu$ gives $|d(tu) - \kappa_u t| \le Ct^2$ with $C = L/(2\|A_0\|_F)$. If $\kappa_u > 0$ and τ is small enough that the roots below lie in $(0, r]$: every t with $\kappa_u t + Ct^2 \le \tau$ has $d(tu) \le \tau$, and every t with $\kappa_u t - Ct^2 > \tau$ has $d(tu) > \tau$; so $t^\star_u$ lies between the smallest positive roots of $\kappa_u t \pm Ct^2 = \tau$, both of which are $\tau/\kappa_u + O(\tau^2)$. (i) $\|A - A_0\|_F^2 = \sum_k \|a_k - a_{0,k}\|^2 = \sum_k d_k^2 \|a_{0,k}\|^2$; divide by $\|A_0\|_F^2$. The same identity applied to $\mathcal{J}u$ row by row gives κ_u^2. (ii) For $t \le s$, $d(tu)/t \ge d(su)/s$, so $d(tu) > \tau$ for every $t \in (\tau s/d(su), s]$, an interval that is nonempty because $d(su) \ge \tau$ gives $\tau s/d(su) \le s$; hence $t^\star_u \le \tau s/d(su)$. Where the expansion holds, a nonincreasing $d(tu)/t$ has its supremum at $t \to 0^+$, which is κ_u. □

By (i) the seven-probe distance is a norm-weighted root mean square of single-probe distances: probing cannot raise the distance above the worst probe, and what it adds is coverage of directions u to which the opening action alone is blind ($\kappa_{u,1} = 0$ while $\kappa_u > 0$). By (ii), when the response is sublinear in the dose ($d(tu)/t$ nonincreasing), the tolerance is bounded by the smallest dose that crosses the threshold, and a linear extrapolation from a large dose, $\tau s/d(su)$ with s large, overstates it.

Check (results/drive_vihan/tolerance/tolerance_openvla_oft.c01e05.jsonl and tolerance_pi05.d4e5bc.jsonl, 20 units each, $K = 1$). The premise $A(0) = A_0$ holds: the logged floor (reference queried twice with the same seed) is exactly 0 on all 40 units. The first-order prediction $d(2t)/d(t) \to 2$ is not reached at the logged doses: the per-unit, per-axis ratio from 0.5 to 1 cm has median 1.46 (OFT) and 1.58 ($\pi_{0.5}$), and log-log slopes of the median curves over doses up to 4 are 0.67 and 0.48 (OFT, cm and degrees) and 0.56 and 0.32 ($\pi_{0.5}$); the linear regime of (31) therefore lies below 0.5 cm and 0.5° if it exists at all, and the probe measures the sublinear part of the response. The medians over units of per-unit axis-medians have $d(t)/t$ nonincreasing on the whole grid in three of four curves (OFT cm: 0.140, 0.104, 0.095, 0.068, 0.040, 0.023 per cm) and on doses up to 8 cm in the fourth; per unit-axis curve it holds on only 10 to 14 of 60, so (ii) is applied to the median curve $m(t)$ in place of $d(tu)$ (its proof uses only that $m(t)/t$ is nonincreasing) and is a statement about the median response. Monotonicity is observed only at the grid doses; between them it is an assumption,

and each bound below uses it on $(\tau s/m(s), s]$ with s the grid dose attaining the minimum. With the frozen $\tau = 0.10$, (ii) gives: OFT $t^\star \le 0.96$ cm and $\le 0.99°$; $\pi_{0.5}$ $t^\star \le 1.39$ cm and $\le 0.47°$, consistent with the grid knees of Fig. 4 (1 cm and 1°; 2 cm and 0.5°). Linear extrapolation from the largest dose would instead give 4.3 cm and 2.6° (OFT), 2.4 cm and 2.8° ($\pi_{0.5}$). Identity (i) holds to 10^{-15} (floating point) on all 230 corrected-frame probe rows of the three files results/drive_vihan/sweep/smc_*.jsonl (OFT 69, $\pi_{0.5}$ 69, OpenVLA 92 rows), with A_0 the raw-frame probe actions of the same task's nominal row; being an algebraic identity it cannot fail, and the check only guards the transcription of (15).

The grid knee as an order statistic†.

Lemma 6 (knee and per-unit tolerances). On a dose grid $t_1 < \cdots < t_G$ let $D_u(t)$ be unit u's distance (the axis-median, as in the knee of Appendix B.1), $m(t) = \text{median}_u D_u(t)$ over n units, $t_u^\star = \min\{t_j : D_u(t_j) > \tau\}$ ($+\infty$ if none), $t^\star_{(1)} \le \cdots \le t^\star_{(n)}$ their order statistics and $\delta^\star = \min\{t_j : m(t_j) > \tau\}$. If every D_u is nondecreasing on the grid, then m is nondecreasing, $\{t_j : m(t_j) > \tau\} = \{t_j : t_j \ge \delta^\star\}$, and $t^\star_{(n/2)} \le \delta^\star \le t^\star_{(n/2+1)}$ for even n (the numpy median averages the two middle values), $\delta^\star = t^\star_{((n+1)/2)}$ for odd n.

Proof. Order statistics of nondecreasing functions are nondecreasing, and so is their mean, so m is nondecreasing and its super-level set is an up-set of the grid. Monotonicity gives $D_u(t) > \tau \iff t \ge t_u^\star$, so $N(t) = \#\{u : D_u(t) > \tau\} = \#\{u : t_u^\star \le t\}$. For even n, $m(t) > \tau$ implies the upper middle value exceeds τ, so $N(t) \ge n/2$, so $t \ge t^\star_{(n/2)}$; and $t \ge t^\star_{(n/2+1)}$ gives $N(t) \ge n/2 + 1$, so both middle values exceed τ and $m(t) > \tau$. For odd n, $m(t) > \tau \iff N(t) \ge (n+1)/2 \iff t \ge t^\star_{((n+1)/2)}$. □

Check (results/drive_vihan/tolerance/tolerance_*.jsonl, 20 units, 3 axes). The premise fails for most units: of the 60 unit-axis curves, 13 (OFT, cm), 13 (OFT, deg), 16 ($\pi_{0.5}$, cm) and 14 ($\pi_{0.5}$, deg) are nondecreasing in the dose, and of the 20 per-unit axis-median curves 8, 8, 9 and 4. The four median curves m are nondecreasing. Where the lemma's conclusion can be compared, OFT (cm and deg: $t^\star_{(10)} = t^\star_{(11)} = 1$, knee 1) and $\pi_{0.5}$ deg ($t^\star_{(10)} = 0.5$, $t^\star_{(11)} = 1$, knee 0.5) agree with it; $\pi_{0.5}$ cm does not (both middle per-unit tolerances are 1 cm, the knee is 2 cm, because $m(1) = 0.099$ sits just under τ while more than half the units have crossed τ at some dose ≤ 1 cm and come back). The grid knee therefore describes the median curve only, not the median unit's tolerance.

Error rates of the motion-free switch†.

Proposition 18 (separation gives the error rates). Model the residual (12) as $r = r_{\text{geo}}(\eta) + e$, where $r_{\text{geo}}(\eta_0) = 0$, the shifts of interest satisfy $r_{\text{geo}}(\eta) \ge \rho > 0$, and the sensing error e is independent of r_{geo} and has the same law under η_0 and under a shift, with mean μ and variance s^2. For the rule "measure iff $r > \tau$" with $\mu < \tau < \rho + \mu$:

$$\Pr(\text{false alarm}) \le \frac{s^2}{s^2 + (\tau - \mu)^2}, \qquad \Pr(\text{miss}) \le \frac{s^2}{s^2 + (\rho + \mu - \tau)^2}. \tag{32}$$

The larger bound is smallest at $\tau^\star = \mu + \rho/2$, where both equal $s^2/(s^2 + \rho^2/4)$. If instead $|e| \le \nu$ almost surely and $\rho > 2\nu$, every $\tau \in [\nu, \rho - \nu)$ has zero error. Finally, if n_0 unshifted and n_1 shifted units are all classified correctly, the exact one-sided $1 - \alpha$ upper confidence limits on the two per-unit error rates are $1 - \alpha^{1/n_0}$ and $1 - \alpha^{1/n_1}$.

Proof. A false alarm is $e > \tau$, i.e. $e - \mu > \tau - \mu > 0$, and Cantelli's inequality $\Pr(X - \mathbb{E}X \ge \lambda) \le \text{Var}X/(\text{Var}X + \lambda^2)$ gives the first bound. A miss is $r_{\text{geo}} + e \le \tau$, which implies $e - \mu \le \tau - \rho - \mu < 0$; Cantelli applied to $-e$ gives the second. The bounds are decreasing in $\tau - \mu$ and in $\rho + \mu - \tau$ respectively, so the larger is minimised where the two arguments are equal, $\tau - \mu = \rho/2$. Under $|e| \le \nu$: unshifted $r \le \nu \le \tau$, shifted $r \ge \rho - \nu > \tau$. With n independent units and error rate ϵ, $\Pr(\text{no error}) = (1 - \epsilon)^n$, which is $\le \alpha$ iff $\epsilon \ge 1 - \alpha^{1/n}$ (the Clopper-Pearson limit for zero events). □

The additive model is the assumption that carries the proposition: the chamfer is not a metric, and the bound holds for the residual only insofar as sensing error adds to the geometric residual rather than interacting with it. The bounds are statements about the population moments; with μ and s estimated from rows they are plug-in values, not confidence statements, and the equal-law and independence assumptions under a shift cannot be checked from unshifted rows (untested).

Check. Graded sweep, simulator masks, one pose (results/local/static_residual.jsonl): the 60 unshifted units (10 nominal, 50 light) read exactly 0, so $\nu = 0$; the 70 camera-shifted units read at least $\rho = 1.929$ (orbit 10°), median 6.45. The zero-error interval is $[0, 1.929)$ and contains $\tau = 1.5$; the 95% limits are $1 - 0.05^{1/60} = 0.049$ and $1 - 0.05^{1/70} = 0.042$. The 170 $\pi_{0.5}$ sweep units of the motion-free variant read at most 0.020. LIBERO-Plus, simulator masks (cal_nominal_score in results/drive_vihan/wraps*/*.jsonl, the pose-averaged statistic S of Eq. 14, not the one-pose r): the 132 unshifted tasks read at most $\nu = 0.015$ and the 191 camera-shifted tasks (distinct task ids) at least $\rho = 3.018$; the 95% limits are 0.022 and 0.016. Learned masks (the full_learned rows of the same files), where the noise floor is real: 21 unshifted rows (all joint-offset conditions) have $\mu = 0.567$, $s = 0.162$ (sample standard deviation), maximum 0.906. Taking $\rho = 3.018$ from the simulator rows, the plug-in values of (32) for statistic S (not the one-pose r; μ and s from the joint-offset rows) at $\tau = 1.5$ are a false-alarm bound of 0.029 and a miss bound of 0.006; $\tau^\star = 2.08$ would give 0.011 for both. The bounded-noise interval with $\nu = 0.906$ is $[0.906, 2.112)$ and contains the post hoc $\tau = 1.5$; the 14 shifted learned rows read at least $4.85 \geq \rho - \nu$, and all 35 learned rows are classified correctly (95% limits 0.133 and 0.193 at these sample sizes). A code docstring written before the current rows ("above 2.26 against below 0.86") refers to an earlier residual scale and is superseded by these values; the sweep's one-pose residual and the LIBERO-Plus multi-pose score differ in scale because the latter averages over poses at which a shifted camera sees the arm from further off.

Clustered zero-event limits for the motion-free switch (adds to Prop. 18)†.

Proposition 19 (zero events in clusters). Let the units fall into G independent clusters, each unit with error probability ϵ and arbitrary dependence within a cluster. If no unit errs, the one-sided $1 - \alpha$ upper confidence limit on ϵ is at most $1 - \alpha^{1/G}$, attained when within each cluster the error event is common (all units err together); so $1 - \alpha^{1/G}$ is a valid limit whatever the dependence within clusters. The unit-level limit $1 - \alpha^{1/n}$ of Prop. 18 is valid only for independent units and is optimistic otherwise.

Proof. For each cluster g pick one unit; the event "no error in g" is contained in "no error at that unit", so it has probability at most $1 - \epsilon$, with equality when the cluster's units err together. By independence across clusters, $\Pr(\text{no error}) \leq (1 - \epsilon)^G$, which is $\leq \alpha$ whenever $\epsilon \geq 1 - \alpha^{1/G}$; so every ϵ above that value is rejected at level α, and the limit is at most $1 - \alpha^{1/G}$. Under the common-event model $\Pr(\text{no error}) = (1 - \epsilon)^G$ exactly and the limit equals $1 - \alpha^{1/G}$. □

Check, $\alpha = 0.05$ (cluster counts from results/local/static_residual.jsonl and the task names of the cal_nominal_score rows in results/drive_vihan/wraps*/*.jsonl; a LIBERO-Plus task id is mapped to its base LIBERO task by stripping the perturbation suffix). Graded sweep: the 60 unshifted and the 70 shifted units each come from $G = 10$ base tasks, so both limits are $1 - 0.05^{1/10} = 0.259$, against the optimistic unit-level 0.049 and 0.042. LIBERO-Plus, simulator masks: the 132 unshifted task ids come from 32 base tasks (limit 0.089, unit-level 0.022) and the 191 shifted ones from 39 (0.074, unit-level 0.016); taking scenes as clusters (the libero_10 scenes plus one scene each for the spatial, object and goal suites, 12 in both groups) gives 0.221. Learned masks: 21 unshifted rows from 7 base tasks (0.348, unit-level 0.133) and 14 shifted rows from 6 (0.393, unit-level 0.193).

The attribution ladder†.

Proposition 20 (well-posedness and misattribution). Let the rungs be nested, each changing the previous rung's frame only by making one more component exact (measured to true

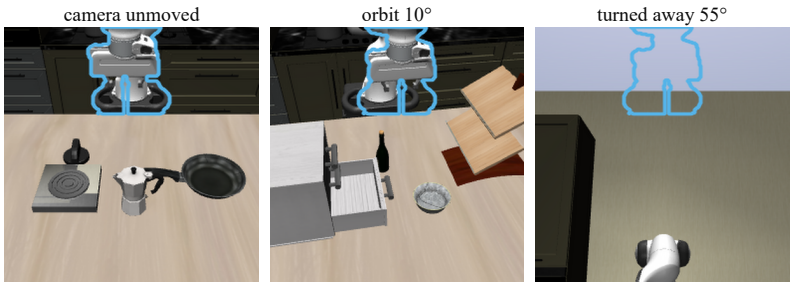


Figure 6: The motion-free test of Prop. 5: the robot's silhouette as the training camera would see it at the current joint state (blue outline) over the deployed frame, from one rest pose with no arm motion. Camera unmoved: it lands on the arm (residual 0.00 px) and the probe motion is skipped. Orbit 10 degrees: it misses the arm by a few pixels (2.08 px, above $\tau = 1.5$), so the robot measures. Turned away 55 degrees: 43.16 px.

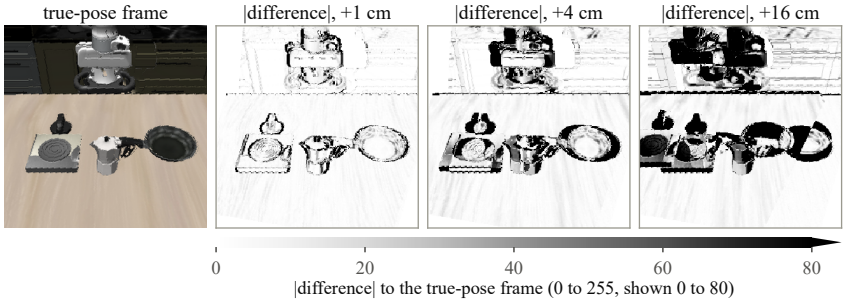


Figure 7: The tolerance in pixels, on one unit of Fig. 4 (task 1167, orbit 10 degrees): the true-pose frame and its absolute difference to the frames at 1, 4 and 16 cm of added position error (mean 7.4, 16.7 and 30.1 on 0 to 255; shown 0 to 80). On this unit the fingerprint distance d at the same doses is 0.078, 0.295 and 0.365 for OFT and 0.062, 0.158 and 0.182 for $\pi_{0.5}$ (family medians 0.104, 0.272, 0.373 and 0.099, 0.246, 0.672). At 1 cm the median opening action has changed by 10% (the reference threshold), while every edge has moved: pixel difference does not measure what the policy tolerates, which is why tolerance is measured on actions†.

pose; reprojection to the training camera's render), and let $Y_k(u)$ be the outcome rung k would have without rollout noise. (a) Rule (17) assigns every raw failure to exactly one bucket, whatever the outcomes. With S_k the number of raw failures that succeed on rung k and n_j the bucket counts, exactly

$$\sum_{j \le k} n_j - S_k = \#\{u : \beta(u) \le k,\ X_k(u) = 0\}, \qquad k = 1, 2, 3, \tag{33}$$

so under monotonicity (M), $Y_1 \le Y_2 \le Y_3$ (making a component exact never turns a noise-free success into a failure), the noise-free buckets are the increments $n_k^* = S_k^* - S_{k-1}^*$ ($S_0^* = 0$, $k \le 3$) and a non-monotone observed pattern certifies at least one flipped rollout. (b) If on every unit each rung's observed outcome differs from Y_k with probability at most f, the observed bucket differs from the noise-free one $\beta^*(u)$ with probability at most $\min(\beta^*(u), 3)\, f$, with no independence assumption, so the expected number of misattributed units is at most $f \sum_u \min(\beta^*(u), 3)$. (c) If in addition the flips are independent across rungs and units, each unit with $\beta^* = 4$ is observed in the policy bucket with probability at least $(1 - f)^3$; if O units are observed there, the one-sided $1 - \alpha$ upper confidence limit on their true number is the largest n with $\Pr\left(\mathrm{Bin}(n, (1 - f)^3) \le O\right) \ge \alpha$.

Proof. (a) β is a minimum over a finite set with a default, hence a function; the buckets are its level sets. For (33): $\sum_{j \le k} n_j = \#\{\beta \le k\}$, and every unit counted in S_k has $\beta \le k$ by

definition of the minimum, so the difference counts exactly the units with $\beta \le k$ that fail rung k. Under (M) and without noise, $X = Y$ is monotone, the right-hand side is 0, and $n_k^* = S_k^* - S_{k-1}^*$; conversely an observed pattern with $X_j = 1$, $X_k = 0$, $j < k$, contradicts (M) for Y, so $X \neq Y$ on some rung. (b) If $X_k = Y_k$ for every $k \le \min(\beta^*, 3)$, then for $\beta^* \le 3$ rungs $1, \dots, \beta^* - 1$ fail and rung β^* succeeds, so $\beta = \beta^*$, and for $\beta^* = 4$ rungs 1 to 3 fail, so $\beta = 4$. The event $\beta \neq \beta^*$ is therefore contained in the union of $\min(\beta^*, 3)$ flip events; the union bound and linearity of expectation give the claims. (c) A unit with $\beta^* = 4$ has $Y_1 = Y_2 = Y_3 = 0$ and is observed as "policy" iff none of the three rungs flips, probability $\ge (1-f)^3$ under independence (across rungs and units); O then stochastically dominates $\mathrm{Bin}(n_{\mathrm{pol}}, (1-f)^3)$, and inverting the binomial tail gives the limit. □

A flip on rung 0 changes which units enter the ladder, not their bucket; it is bounded by the same f per unit. For a deterministic policy a "flip" is the sensitivity of the outcome to the small state and observation differences between rollouts, which the proposition models as random.

Check (attribution_ladder.units on results/drive_vihan/wrapper_oft/rows.jsonl and results/drive_vihan/rows_oft.jsonl, set A, all rungs on all tasks). Success counts 13, 19, 21, 27 and buckets $n = (7, 2, 8, 0)$ reproduce Table 3. Among the 17 raw failures $S = (7, 8, 14)$ against cumulative buckets $(7, 9, 17)$; the differences $0, 1, 3$ are exactly the non-monotone units of (33) (task 803: pattern $0, 1, 0, 0$; 800: $0, 1, 1, 0$; 1065: $0, 0, 1, 0$). One raw success (task 889) is broken by the measured rung and restored by the upper rungs; on the other four OFT sets the upper rungs were run only on raw failures, so no raw success is broken there by them, and the measured rung breaks 17 more (18 of 140 in all). For f we use the only single-rollout flip rate measured on this family: OFT sweep episodes whose outcome changed between raw and wrapped with nothing applied, 7 of 53 (results/drive_vihan/sweep/episodes_openvla_oft*.jsonl, standard conditions), $\hat{f} = 0.132$, one-sided 95% limit 0.234; it includes the sweep's arm motion, which all of rungs 1 to 3 share, so it overstates the noise between rungs; and a raw/wrapped discordance of two rollouts that each flip with probability $f \le 1/2$ occurs with probability $2f(1-f) \ge f$, so $\hat{f}$ overstates f again. Two caveats: (b) needs a bound f on every unit, and $\hat{f}$ is an average over a different task set (the sweep's base tasks); and the sums below are evaluated at the observed buckets in place of the noise-free ones, so they are plug-in estimates, not bounds. Plugging the observed buckets into (b): at most $0.132 \times 35 = 4.6$ of the 17 units are misattributed in expectation on set A, and at most $0.132 \times 161 = 21$ of the 100 pooled units (buckets 61, 17, 21, 1, so $61 \cdot 1 + 17 \cdot 2 + (21 + 1) \cdot 3 = 161$). The bucket shares are therefore uncertain at the level of tens of units; the policy bucket is not: with $O = 1$ observed of 100, (c) gives at most 5 true policy failures at $\hat{f}$ and 8 at its upper limit (one-sided 95% limits given f; the uncertainty in f is covered only by using its upper limit, so by a union bound the statement "at most 8" holds at joint one-sided level at least 90%, assuming the sweep's f bounds f on every ladder unit). Of the pooled calibration errors (Fig. 8b), 13 of the 14 only-true-pose OFT units with a logged error exceed 2 cm (median 39.4 cm); of the 19 only-render OFT units with a logged error, 9 are within 2 cm and 10 exceed it (6 exceed 20 cm; median 2.2 cm); of the 51 OFT units first repaired at the measured pose with a logged error, 19 exceed 2 cm and 10 exceed 20 cm (errors: cam_est_err_cm_deg of the full row, joined to the buckets of Eq. 17).

C Real images: DROID

Fig. 9 reports the self-calibration on real images. Offline calibration of DROID from robot masks is not new (Xie et al., 2025; Irshad et al., 2025). What we add is why the obvious pipeline fails and what repairs it: completing a mask with frame-to-frame motion re-attaches moving background unless motion is trusted only inside regions already called robot (precision 0.91 falls to 0.75 otherwise), and keeping the largest connected component drops parts of an arm cut by the frame edge (recall 0.68 to 0.53 on one camera). The ground-truth-free self-check, the render IoU of the estimated camera, correlates with the true position error at $\rho_s = -0.40$ ($p = 0.003$, $n = 53$)†; it was chosen after the data among three logged statistics (the calibration score at the estimate, +0.37; the mask quality, −0.14). A dedicated robot

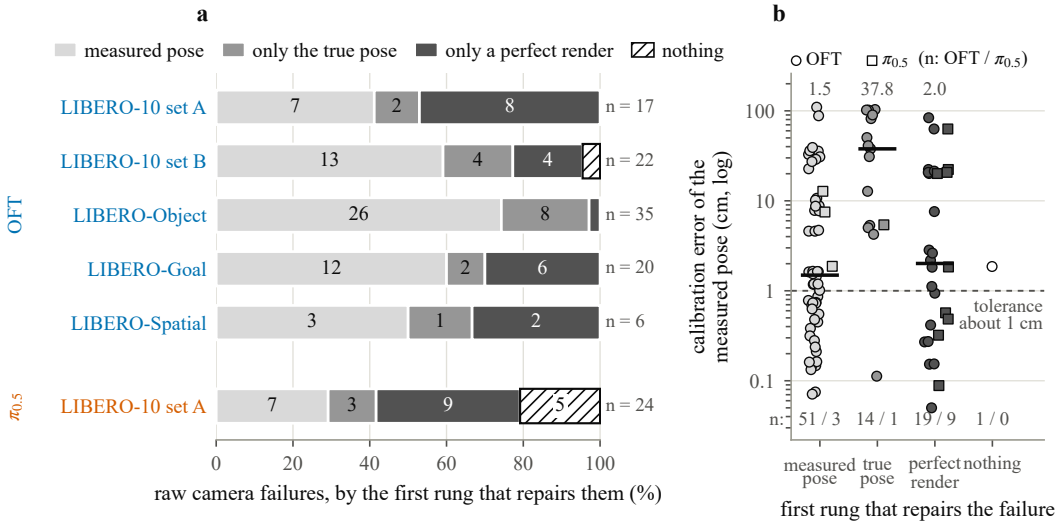


Figure 8: (a) Every raw camera failure attributed to the first rung that repairs it (first repaired at the measured pose; only the true pose; only a perfect render; nothing; light to dark grey, hatched), per source; rows 1 to 5 are OFT, row 1 is set A and repeats Table 3. Pooled over the five OFT rows, 61, 17, 21 and 1 of 100; on the held-out $\pi_{0.5}$, 7, 3, 9 and 5 of 24. (b) Calibration error of the measured pose per bucket (OFT circles, $\pi_{0.5}$ squares; n per strip, OFT / $\pi_{0.5}$: 51 / 3, 14 / 1, 19 / 9, 1 / 0, 98 units with the error logged; bars and numbers above: medians, none for a single unit; dashed: 1 cm, the action-level tolerance on mild orbits†): units only the true pose repairs are gross calibration failures (median 37.8 cm; OFT alone 39.4 cm); units only a render repairs have median error 2.0 cm (OFT alone 2.2 cm, but 10 of its 19 exceed 2 cm).

segmenter (Mei et al., 2025) would be a natural third mask source; the comparison here holds the search fixed and varies only the mask.

D Additional results and figures

Score: first test and task-cluster intervals. The first score test used three base tasks per family (Table 4). Task-cluster bootstrap intervals for Table 1: OFT [0.73, 0.92], $\pi_{0.5}$ [0.66, 0.94], OpenVLA [0.59, 0.82] (3 solved base tasks, so its cluster interval is unreliable). Against the category rate fit on OFT, the paired score difference on $\pi_{0.5}$ is +0.109 (unit bootstrap [−0.03, 0.24], task-cluster [−0.03, 0.25]) and on OpenVLA +0.061 [−0.10, 0.22]; against the label +0.294 [0.18, 0.39]; against ident alone +0.183 [0.09, 0.30]; against invert alone −0.010 [−0.10, 0.07]. A finer baseline†, the repair rate per condition fit on OFT, reaches 0.745 on $\pi_{0.5}$, and the score minus it is +0.077 [−0.02, 0.19]: the score is not shown to be the best category-based predictor, only numerically above both. Shift magnitude, where defined, gives 0.29 (translation, $n = 45$) and 0.82 (rotation, $n = 45$) on $\pi_{0.5}$. The look-away units are shown against the score in Fig. 10, and Lemma 5 splits the difference between the score and invert alone into the pairs that ident moves. Units are (base task, condition), one rollout each (score_baselines.py; no key repeats in the sweep rows, so no averaging occurs). Across categories ident gains about 0.04 by demoting negatives and gives back as much on repaired units, so the pooled score ties invert alone. On look-away units ident is below 1 on 11 of 18 on $\pi_{0.5}$ and OFT (3 of 6 on OpenVLA), its net pair contribution there is +0.018 on OFT and −0.005 on $\pi_{0.5}$, and OFT repairs 2 of 18, OpenVLA 0 of 6. A registered rule that corrected the camera only at coverage ≥ 0.5 failed its pre-stated criterion (2 breaks prevented, 3 repairs lost; App. A).

The deployable score†. (score_selfmeasured_analysis.py, selfmeasured_gate.py; units of Table 1; non-camera units unchanged; task-cluster intervals.) AUROC on OFT, $\pi_{0.5}$, Open-

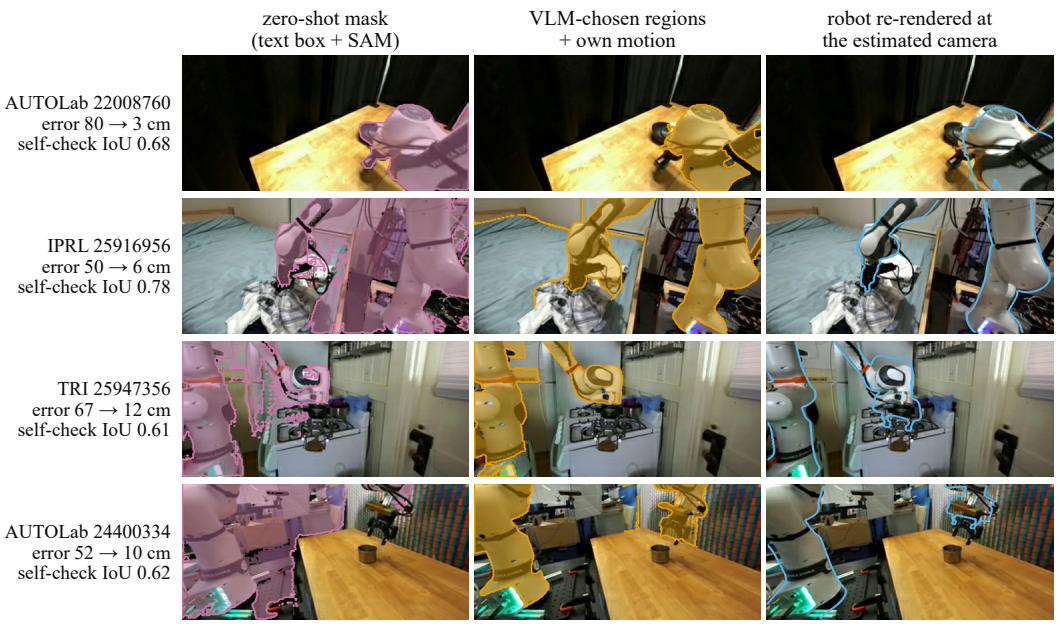


Figure 9: Real DROID frames, 4 of the 53 cameras (chosen to show one camera per failure mode of the zero-shot mask), with the calibration error before and after and the ground-truth-free self-check IoU per row. Left: a zero-shot text-box-plus-SAM mask (purple) attaches background to the robot. Middle: SAM regions chosen by a vision-language model (orange), with part-robot regions split by the robot's own motion. Right: the robot re-rendered at the estimated camera (blue outline). Over all 53 cameras, with identical search and only the mask changed, the median error is 25.4 cm / 25.1° zero-shot against 19.6 cm / 19.3° VLM-chosen; under 30 cm 28 against 40, under 20 cm 19 against 28 of 53 (paired Wilcoxon $p = 0.015$; the mask was changed after the registered stop rule had fired, so the change is post hoc).

VLA and within-category on the pooled held-out families: registered (true-pose ident) 0.823, 0.809, 0.675, 0.737 [0.64, 0.87]; true-pose ident recomputed in the rescoring run 0.826, 0.805, 0.671, 0.730; self-measured ident, no gate 0.765, 0.773, 0.666, 0.679 [0.57, 0.81]; self-measured ident with the residual gate 0.802 [0.68, 0.91], 0.837 [0.73, 0.93], 0.691 [0.62, 0.82], 0.713 [0.60, 0.84]. Calibration failure (> 2 cm) on the 119 camera units with a logged calibration: the residual ρ has AUROC 1.000 on each family; the gate flags 12 of 12 OFT, 36 of 36 $\pi_{0.5}$ (one false alarm of 34) and 8 of 8 OpenVLA failures (none of 20). The self-check F1 is logged on 53 of the 119 units and is weaker (0.614). Under learned masks and monocular metric depth (set A, 30 units) the same τ separates the one fit within 2 cm from all 29 beyond. On real DROID images (53 calibrations, VLM-chosen masks; droid_residual_gate.py) the residual is weaker: it ranks calibrations beyond 10 cm at AUROC 0.71 [0.50, 0.88] and beyond 20 cm at 0.60 [0.44, 0.76], so the gate is not shown to transfer to real images.

Within-category ranking†. (wrapper/scripts/within_category.py; same units; cluster bootstrap over base tasks, 2,000 resamples, seed 0.) A predictor constant within a stratum has AUROC exactly $\frac{1}{2}$ over (repaired, unrepaired) pairs drawn from the same stratum, so the stratified AUROC $A_{\text{strat}}(s) = \sum_g \#\{(i,j) \in g : i \text{ repaired}, j \text{ not}, s_i > s_j\}/\#\{\text{such pairs}\}$ (ties one half) measures only what a predictor adds on top of any category router. Over same-category pairs (OFT 366, $\pi_{0.5}$ 502, OpenVLA 132) the score reaches 0.794 [0.62, 0.94], 0.854 [0.75, 0.96] and 0.720 [0.50, 0.85], and 0.737 [0.64, 0.87] pooled over the held-out families; invert alone 0.709, 0.761, 0.693 (0.684 pooled); ident alone 0.661, 0.656, 0.614. Paired within-category differences (cluster bootstrap): score minus ident +0.151 pooled held-out (one-sided $p = 0.001$), significant on every family except OpenVLA; score minus invert +0.093 on $\pi_{0.5}$ ($p = 0.035$), +0.053 pooled held-out ([−0.02, +0.14], $p = 0.079$), +0.027 on

OpenVLA (n.s.). Within a single condition there are too few pairs (148 pooled held-out) to conclude anything.

Stronger baselines†. (score_baselines.py--ext; unit bootstrap, 2,000 resamples, seed 0; paired differences on the same units.) min(ident, invert) matches the product on every family (OFT 0.828 against 0.823; $\pi_{0.5}$ 0.795 against 0.809; OpenVLA 0.678 against 0.675; paired differences within ±0.04; App. B, "Why a product"). On $\pi_{0.5}$ camera units the true rotation magnitude reaches 0.818 [0.68, 0.93] against the score's 0.877 on the same 45 units (score minus it +0.059 [−0.05, 0.18]): it ties the score. True-pose coverage and a true-pose image statistic are logged only policy-free (results/local/fidelity.jsonl), on the three envelope base tasks, which leaves 12 to 18 units per family (OFT: true-pose coverage 0.806 against the score's 0.861 on the same 12 units), too few to rank. No image statistic of the corrected frame the wrapper actually fed the policy (pixel error or SSIM against the training reference) is logged in any rollout row, so that baseline cannot be computed from the rows; logging it at the first policy query is a Colab item. On the 98 in-rollout camera failures of Fig. 2b, coverage beats translation (paired +0.222 [0.07, 0.38]) and is numerically above rotation (+0.085 [−0.01, 0.19]; coverage_paired.py, same bootstrap).

Why the weaker family scores lower†. On OpenVLA, 5 of its 24 repaired units had no channel applied (OFT: 0 of 37; $\pi_{0.5}$: 1 of 37): the measurement's arm motion jostled a fragile policy into success, and four of the five are background units at recover = 0. With P repaired units, m of them of this kind, and $\widehat{A}_{\mathcal{G}}$, $\widehat{A}_{\mathcal{C}}$ the AUROC on the other and on these positives, the AUROC is exactly $\widehat{A} = \frac{P-m}{P}\widehat{A}_{\mathcal{G}} + \frac{m}{P}\widehat{A}_{\mathcal{C}}$ (Prop. 16), and these units take 0.816 to 0.675, to the digit. Redefining the outcome symmetrically, repaired$^*(u)$ = repaired$(u) \wedge$ (some channel applied), and dropping every unit where nothing was applied, gives 0.817 ($n = 64$), 0.826 (70) and 0.782 (49) (one-sided exclusion: 0.823, 0.831, 0.816): the gap to the best family shrinks from 0.15 (0.823 vs 0.675) to 0.04 (0.826 vs 0.782). The registered result stays 0.675.

Ladder buckets: uncertainty and error. Bucket shares are uncertain at the level of about 21 of 100 units (App. B.4); the bound of at most 8 policy units holds at joint one-sided level at least 90% (95% for f, 95% given f) and is a plug-in estimate. Only-render units have median calibration error 2.2 cm, but 10 of 19 logged exceed 2 cm, and they have the lowest self-measured coverage (render-only against the other two buckets, AUROC 0.81 [0.65, 0.94], task-cluster, $n = 97^{\dagger}$). Of the 51 logged measured-rung repairs, 19 exceed 2 cm and 10 exceed 20 cm, so those repairs are not due to reprojection (fill or arm motion) and fall inside the misattribution budget. The light channel on $\pi_{0.5}$ shows a harmful trend (3 repaired, 13 broken over three post hoc replicates; task-level $p = 0.078)^{\dagger}$; the $\pi_{0.5}$ camera replicates give sign test $p = 0.035$ and exact sign-flip $p = 0.12$ on the task-level summed difference.

The cost of the measuring motion. The probe motion hands the arm back 10 to 25 mm from where the episode began and still moving at 0.19 rad/s; re-running the raw policy with the same environment and sampling seeds changes 0 of 20 outcomes (one-sided 95% upper bound 14%; layout and background tasks), against 15 of 48 changed when the arm moved and nothing else was applied.

Probe triangulation and divergence. The triangulation residual (the root-mean-square angle between each probe's commanded direction and the direction to the least-squares meeting point) does not predict wrapped success on OFT or $\pi_{0.5}$ (AUROC 0.32 to 0.56). A linear field with $\operatorname{tr} A > 0$ has no attracting sub-goal (Prop. 6), and the fitted field has $\operatorname{tr} \hat{A} > 0$ on 97%, 97% and 96% of units (OFT, $\pi_{0.5}$, OpenVLA; above 2 standard errors on 3%, 36% and 26%): the fitted local field expands volume on the probed region, whose equilibrium lies outside the 10 cm probe ball on 220 of 240 units, so it says nothing about an attractor (Prop. 15). Positive divergence is significant at unit level ($p < 10^{-18}$ in each family) but rests on 3 to 4 base tasks (task-level $p = 0.125, 0.125, 0.0625$; Lemma 3).

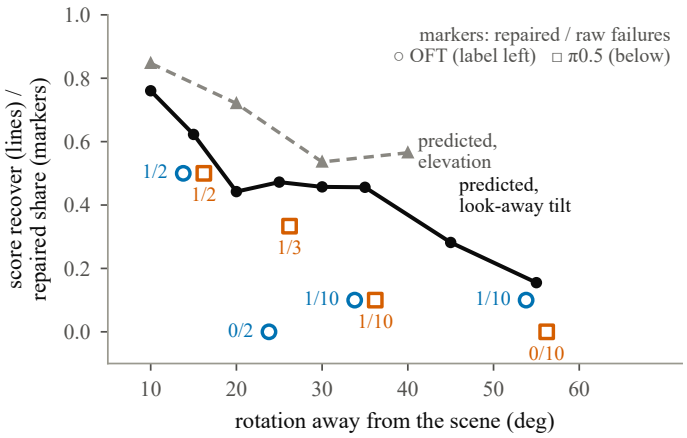


Figure 10: The score as the camera rotates away from the scene (black: look-away tilt; grey dashed: elevation; mean over three base tasks). The policy-free map scores a 35-degree look-away 0.46 on average, while the sweep's own scores at that condition, from the self-measured pose, average 0.24 over its 10 base tasks. Markers: repaired / raw failures of the graded sweep at each look-away condition, all ten base tasks (not the Table 1 prediction units), OFT left of and $\pi_{0.5}$ below each marker.

Outcome tolerance in rollouts†. (wrapper/scripts/outcome_tolerance.py.) On the 98 in-rollout camera raw failures of Fig. 2b (85 OFT, 13 $\pi_{0.5}$, all with the pose applied; 54 repaired), the repair rate by calibration error is 22 of 34 (0.65 [0.48, 0.79], Wilson) below 1 cm, 11 of 15 (0.73 [0.48, 0.89]) at 1 to 2 cm, 3 of 7 (0.43 [0.16, 0.75]) at 2 to 5 cm, 8 of 13 (0.62 [0.36, 0.82]) at 5 to 20 cm and 10 of 29 (0.34 [0.20, 0.53]) at 20 cm or more. The isotonic (non-increasing) fit is flat at 0.71 up to 1.83 cm, then 0.50, 0.47 and 0.17 beyond 41 cm (breakpoints 1.83, 20.1, 41.0 cm): in rollouts there is no knee at 1 cm, and the action-level tolerance of Fig. 4 is a statement about opening actions, not outcomes.

Table 4: The first score test, three base tasks per family (the setting in which the score was first frozen).

	units	repaired	AUROC, score	AUROC, label
OFT (development)	21	14	0.908 [0.68, 1.00]	0.500
$\pi_{0.5}$ (held out)	21	14	0.837 [0.64, 1.00]	0.464
pooled	42	28	0.881 [0.76, 0.98]	0.482

Baselines on set B. Table 5 runs every method on every task of set B. Test-time augmentation (TTA) breaks nothing, but over 150 camera tasks on three suites the wrapper beats it on 46 tasks and loses on 13 ($p < 10^{-4}$), and TTA alone does not beat the raw policy on LIBERO-Object or LIBERO-Goal.

Table 5: Baselines on set B, the 30 LIBERO-10 camera tasks after the first 30 (OFT; 22 raw failures, 8 raw successes), every method on every task; disjoint from set A (Table 3). The measured pose without depth, generic fixes and test-time augmentation all lose to the wrapper.

method	success	repaired	broken	wrapper wins / losses, p
raw policy	0.27			
generic fixes (WB, CLAHE, denoise)	0.37	3/22	0/8	10 / 2, 0.039
homography, no depth	0.40	9/22	5/8	9 / 2, 0.065
TTA	0.47	6/22	0/8	8 / 3, 0.23
wrapper	0.63	13/22	2/8	

Level-5 tasks. On LIBERO-Plus level-5 tasks (the hardest grade), the servo repairs 32 of 33 robot-initial-state failures, and coverage still ranks level-5 camera failures (AUROC 0.75, $n = 47$).

Table 6: Secondary paired tests and Holm-adjusted p over the $m = 15$ secondary tests, one row per test, ordered by raw p (step-down, monotone; holm_recompute.py). The three task-level rows use the exact sign-flip test (Eq. 16): the motion-free row over the 7 tasks with a non-zero difference (the episode-level McNemar is 0.0125, Holm 0.088 in that family), the two replicate rows over 3 replicates; all other rows use the exact McNemar test (Eq. 3), or Wilcoxon where stated. Primary: pooled OFT camera, 61 of 100 repaired, 18 of 140 broken, $p = 1.3 \times 10^{-6}$.

comparison	repaired / broken	p	Holm p
OFT joint, three untuned suites	47 / 1	3.5×10^{-13}	5.2×10^{-12}
OFT Object joint	16 / 0	3.1×10^{-5}	4.3×10^{-4}
OFT Goal joint	16 / 0	3.1×10^{-5}	4.3×10^{-4}
$\pi_{0.5}$ joint	16 / 0	3.1×10^{-5}	4.3×10^{-4}
OFT LIBERO-10 joint	18 / 1	7.6×10^{-5}	8.4×10^{-4}
OFT LIBERO-10 camera	20 / 3	4.9×10^{-4}	0.0049
OFT Spatial joint	15 / 1	5.2×10^{-4}	0.0049
OFT Object camera	26 / 6	5.4×10^{-4}	0.0049
DROID masks (Wilcoxon)†		0.015	0.11
motion-free vs. raw, $\pi_{0.5}$ sweep, task level†	21 / 7	0.047	0.28
$\pi_{0.5}$ light, 3 replicates, task level†	3 / 13	0.078	0.39
$\pi_{0.5}$ camera, 3 replicates, task level†	21 / 8	0.12	0.48
OFT Goal camera	12 / 5	0.14	0.48
$\pi_{0.5}$ camera, one rollout	7 / 2	0.18	0.48
OFT Spatial camera	3 / 4	1	1

The action-field fingerprint. Table 7 gives the fingerprint ablations and selection results (69 units per family; AUROC with unit-bootstrap intervals, from wrapper/scripts/fingerprint_ablation.py). The probes carry the signal on the two coherent families: seven probes reach 0.93 and 0.79 against 0.57 and 0.68 for the opening action alone, and on OFT the AUROC rises with the number of probes. OpenVLA never matches its own training reference (none of its units comes within the frozen 0.10 of it), so the statistic has nothing to compare against. Across the registered decision rules, predicting what is repairable proves easier than deciding frame by frame when to repair: a do-no-harm veto and a coverage threshold also did no better than always correcting (Appendix A).

Table 7: Action-field fingerprint; constants frozen on OFT before any held-out probe; 69 units per family. The selector beats always correcting only on $\pi_{0.5}$ (0.86 against 0.81); H-F1 fails on OpenVLA.

	OFT (dev.)	$\pi_{0.5}$ (held out)	OpenVLA (held out)
AUROC, 7 probes	0.928 [0.85, 0.99]	0.787 [0.61, 0.92]	0.643 [0.49, 0.77]
AUROC, 3 probes†	0.843 [0.70, 0.97]	0.802 [0.63, 0.93]	0.701 [0.57, 0.82]
AUROC, one probe (centre)†	0.801 [0.64, 0.94]	0.751 [0.62, 0.86]	0.693 [0.56, 0.81]
AUROC, opening action only†	0.571 [0.42, 0.71]	0.679 [0.51, 0.84]	0.465 [0.32, 0.61]
success, always raw	0.70	0.70	0.13
success, always wrapped	0.84	0.81	0.46
success, fingerprint selects	0.80	0.86	0.32
success, per-unit oracle	0.90	0.90	0.48
units a correction broke, kept raw	0 of 1	5 of 5	0 of 1

The self-check extension†. On the 150 camera tasks where it was run (LIBERO-10 set B, Object, Goal), full_verify (settle, self-check, re-measure with seven poses when F1 < 0.75)

repairs 55 and breaks 10 against raw ($p = 1.2 \times 10^{-8}$) and beats full on 19 tasks against 12 ($p = 0.28$); success on set B rises from 0.633 to 0.733. The 31 discordant tasks are what single-rollout noise alone predicts: two rollouts that each flip with probability f disagree with probability $2f(1-f)$, and $2(0.132)(0.868) \times 150 = 34$ at the OFT flip rate. On OFT camera tasks the measure-then-settle variant beats measuring alone on 7 tasks and loses on 2 over 90 ($p = 0.18$); its 9 discordant tasks are fewer than the 21 that noise alone predicts, so neither comparison separates the variants from rollout noise. The self-check fired on 31 of the 150 tasks, and the re-measured calibration was kept (higher F1) on 16. It repairs 6 of the 8 Object units the ladder attributes to calibration, but the second measurement rarely fixed the pose (errors after it 5 to 97 cm), so the mechanism is not claimed.

The motion-free variant, decomposed†. On the 40 non-camera $\pi_{0.5}$ tasks of Fig. 3a (results/drive_vihan/wraps/wrap_pi05.jsonl, one rollout each, wrapper/scripts/motionfree_tests.py) the success rates are 0.75 (raw), 0.55 (measure always, full), 0.675 (measure then settle) and 0.70 (motion-free check first), and

$$\mathrm{SR}_{\mathrm{check}} - \mathrm{SR}_{\mathrm{full}} = (\mathrm{SR}_{\mathrm{check}} - \mathrm{SR}_{\mathrm{settle}}) + (\mathrm{SR}_{\mathrm{settle}} - \mathrm{SR}_{\mathrm{full}}) = (+0.025) + (+0.125) = +0.150. \tag{34}$$

Exact McNemar on discordant tasks: check vs. settle 6 / 5, $p = 1.0$; settle vs. full 6 / 1, $p = 0.125$; check vs. full 8 / 2, $p = 0.11$; check vs. raw 1 / 3, $p = 0.63$. Settling the arm after the probe motion supplies most of the gain, and on these tasks the check adds nothing measurable over settling. On the graded sweep (170 non-camera units, 10 base tasks, all $\pi_{0.5}$ shards merged) the per-task differences, check minus measure always, are $D = (3, 10, 0, 4, 0, 6, 0, 2, 4, 0)$: $n_0 = 6$, null size $2^6 = 64$, smallest attainable $p = 0.031$, and the exact sign-flip test (16), the sign test and Wilcoxon all give $p = 0.031$ (episode-level McNemar 41 / 12, $p = 8.2 \times 10^{-5}$). Check minus raw: $D = (0, -1, 0, 1, 3, 0, 3, 2, 2, 4)$, $n_0 = 7$, $2^7 = 128$, sign-flip $p = 6/128 = 0.047$, Wilcoxon 0.047, sign test 0.125 (episode-level 21 / 7, $p = 0.013$). Measure always minus raw: sign-flip $p = 0.38$. On these units the check never fires, so it is equivalent to never measuring, and its ability to fire rests on the policy-free separation of Fig. 3b. The largest static residual over the 170 sweep units is 0.020 px (joint offset 0.20 on task 1442; next 0.019, offset 0.35 on task 1167); 145 of 170 are exactly 0, and all 25 nonzero values are on units where the arm was servoed to the training start just before the check (11 joint-offset units and the 14 other units of task 1101, whose start is 0.071 rad off); the model mask is rendered at the encoder reading, not at a stored rest pose (App. B).

Rollout noise on $\pi_{0.5}$. The three replicated runs of Sec. 4.2 hold the task and the perturbation fixed and vary the seed (environment and policy sampling seed = task id $+1000\times$ replicate). Across them the outcome of a task changes on 10% of camera tasks under the raw policy and 33% under the measuring wrapper, the arm-motion mechanism of Sec. 4.3 seen as variance; with the same seed the raw policy changes 0 of 20 outcomes. On the 10 light tasks run alongside, the motion-free variant breaks 1 of 9 raw successes where measuring always breaks 3 of 9; over the three replicates the measuring wrapper breaks 13 and repairs 3 episodes (episode-level $p = 0.021$). Task-level (wrapper/scripts/replicate_tests.py), with $D_i \in \{-3, \ldots, 3\}$ the summed wrapped-minus-raw difference over the three replicates: camera, 30 tasks, histogram $\{-3{:}\,2,\ -1{:}\,1,\ 0{:}\,15,\ 1{:}\,6,\ 2{:}\,4,\ 3{:}\,2\}$, so 12 positive, 3 negative, $n_0 = 15$, null size $2^{15} = 32{,}768$, smallest attainable $p = 6.1 \times 10^{-5}$; sign test $p = 0.035$, sign-flip $p = 0.119$, Wilcoxon $p = 0.094$. Light, 10 tasks, $\{-3{:}\,2,\ -2{:}\,1,\ -1{:}\,3,\ 0{:}\,3,\ 1{:}\,1\}$, 1 positive, 6 negative, $n_0 = 7$, $2^7 = 128$, smallest $p = 0.016$; sign $p = 0.125$, sign-flip $p = 0.078$, Wilcoxon $p = 0.078$. The camera tests differ because the sign test counts only the 12 against 3 signs, while the sign-flip test weights each task by $|D_i|$ and the two tasks at -3 weigh heavily against a positive side made mostly of $+1$s; Wilcoxon ranks the magnitudes and falls between the two.

A nuisance the body does not model. Zooming the true camera while the wrapper assumes the training focal length (8 camera tasks, policy-free) is absorbed as a camera that backed away along its own optical axis: at 5, 10 and 20% zoom the pose error is 2.7, 5.9 and 10.8 cm, almost all of it along that axis. The corrected frame degrades gracefully (image error 14.5, 16.3 and 20.1 against about 30 uncorrected and 10.2 with the true focal length), and

the self-check barely notices (F1 0.93 to 0.83): it checks the pose it was asked about, not the model behind it. Focal length is not aliased with the 6-DoF pose (Jacobian residual 0.78 against a median of 0.51 among the pose columns), so three poses at different depths identify it; lens shift in x is aliased with a 0.44 degree pan, which undoes it.

Figures. Fig. 8 gives the ladder per source and the calibration error per bucket; rows 3 and 4 of Fig. 1 show the two upper rungs in pictures. Fig. 10 gives the look-away line. Fig. 5 tests Props. 4 and 12; Fig. 6 shows the motion-free test and Fig. 7 the frames behind the tolerance probe. Over the 88 camera placements $\times$ 3 base tasks of Fig. 2a (results/local/envelope.jsonl, ident $= \sigma_{\min}/\sigma_{\min,\mathrm{ref}}$ capped at 1), ident ≥ 0.9 on 199 of the 264 task-placements (at least 0.61 on all 240 that do not turn away from the scene; minimum 0.18 at a 55 degree look-away), so across the orbit and distance map recover tracks coverage (Pearson $r = 0.84$ with coverage and 0.03 with identifiability over the 192 task-placements of the azimuth-by-distance grid, App. B.1).